

TRIAD

Air Combat Simulator — the complete history of every request

From Sat Sep 26, 2026 · 00:11 UTC to Sun Sep 27, 2026 · 23:45 UTC

91 requests · 39 released versions · 58 commits · 37 reference images shared

This document collects every message you sent while building TRIAD, word for word and in order, followed by the release notes for every version the game shipped and a snapshot of what the game is today. Nothing is paraphrased in Part 1: the requests are exactly as typed (images are noted but not reproduced).

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At a glance: the road to v4.5.1

Version	Date	What shipped
v1.0.0	2026-09-26	First release
v1.1.0	2026-09-26	High-detail aircraft
v1.2.0	2026-09-26	Realistic flight physics
v1.2.1	2026-09-26	Smooth roll-outs
v1.3.0	2026-09-26	Auto-Fly
v1.4.0	2026-09-26	5v5 Team Battle
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v1.9.0	2026-09-26	Frostfall Strait, new water, 3 new wraps
v1.9.1	2026-09-26	Giant mountains
v1.9.2	2026-09-26	Chase camera auto-recenter
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v4.0.0	2026-09-27	Multiplayer
v4.0.1	2026-09-27	Intakes the right way up
v4.1.0	2026-09-27	The Rafale is coming
v4.2.0	2026-09-27	Hero-detail jets
v4.3.0	2026-09-27	Weather
v4.3.1	2026-09-27	Rain on the canopy
v4.3.2	2026-09-27	Smaller rain drops

Version	Date	What shipped
v4.4.0	2026-09-27	Gun lead marker
v4.4.1	2026-09-28	Fixes: clouds, reflections, weather panel
v4.5.0	2026-09-28	F-22A Raptor
v4.5.1	2026-09-28	F-22A model rebuilt

Part 1 — Every request, word for word

Numbered in the order you sent them. Times are UTC. Where you attached screenshots or reference photos, the number of images is noted.

Saturday, September 26, 2026

#1 · 00:11 UTC · 5 images attached

have it make DCS clone BUT. The only Vehicles it should make is F-15 EX II, F-18 Super Hornet, and Eurofighter, NO OTHER VEHICLES, NO TANKS, NO BOATS, NO OTHER JETS EXCEPT THE ONE I SAID.

no, claude is makking it, you are making the prompt, and claude is going to make like, around 30,000-60,000 lines of code. the game must have a MASSIVE map because the jets are flying at like mach f15 is Mach 2.5, f18 is 1.8, and Eurofighter is mach 2. , so map needs to be HUGE. each jet has their own set of weapons, the map contains trees scattered around everywhere, mountains and hill everywhere, no grass, lets say 6 (3 are enemies, 3 are the users team) airports are scattered evenly across the map, the map is 400x400 nautical miles, the jets have fuel, using afterburners decreases fuel at a very fast rate, but the pro is fast speed. (for all jets). the game modes are free flight, just the user flying in their choice of jet. the next is waves, npc jets get more advanced after each wave is compelted by the user, 10 waves total, wave 1-3 is 3 jets. wave 4-6 is 6 jets, wave 7-9 is 6 jets, wave 10 is 9 jets, and remember after each wave they get more advanced

F-15EX Eagle II specs:

Top speed Mach 2.5

Crew is 2

Length is 63.8 Feet

Wingspan is 42.8

Height is 18.5

Maximum Takeoff Weight is 81,000LBS

Powerplant is Two General Electric F110-GE-129 Turbofans each providing up to 29,500LBS of thrust with afterburners

Combat Range (how far it can fly): 687 Nautical Miles

Service Ceiling is 60,000

Positive G-Forces (Pulling Up / Tight Turns) When the pilot pulls the flight stick back, blood drains from their head to their boots. The screen changes dynamically to show them losing consciousness: 4.0G to 7.9G (The Grey-Out): The camera's color begins to drain. The edges of the screen fade into a blurry, desaturated gray as the pilot starts losing peripheral vision. 8.0G to 10.4G (The Monochrome Tunnel): The screen turns completely black and white. The player is experiencing severe tunnel vision and is on the verge of passing out. 10.5G or Higher (G-LOC Blackout): The pilot completely passes out (G-Induced Loss of Consciousness). The screen goes pitch black for exactly 10 seconds. During this time, the controls completely freeze—if the jet was heading toward a mountain when the player blacked out, it will likely crash before they wake up. When the pilot pushes the flight stick forward, blood rushes intensely into their head and eyes, causing a dangerous condition called a Redout: -2.0G to -4.9G (Partial Redout): A crimson vignette creeps into the borders of the screen, creating a 50% solid red tint over the player's view. -5.0G or Lower (Full Redout): The screen turns 100% solid, saturated red. The player's vision is completely blinded by blood pressure in the optic nerves, making it impossible to see the HUD or the terrain.

Armament Payload is 23 hardpoints supporting up to 29,000LBS of ordinance (including up to 12 air to air missiles) plus an internal M61A1 20mm rotary cannon

missiles and bombs supports AIM-120D and AIM-9X

Avionics Systems

Radar AN/APG-82(V)1 Active electronically scanned array (aesa) radar

**electronic warfare eagle passive active warning survivability system (EPAWSS)

Flight control is Digital fly by wire system

F/A-18E/F Super Hornet Specs:**

Length is 60.3 feet

wingspan is 44.9 feet

height is 16 feet

crew is 2

Performance

max speed is mach 1.8

service ceiling is 50,000ft

combat range is 1,275 nautical miles

Engines and weight

Two General Electric F414-GE-400 afterburning turbofan engines

Thrust 22,000LBS per engine With afterburner

max takeoff weight is 66,000LBS

Armament

Cannon is Internal M61A2 Vulcan 20mm rotary cannon

External weapon stations (hardpoints) is 11

missiles and bombs supports AIM-120D and AIM-9X

Eurofighter Typhoon

Crew is 1 seat

length is 52.4 feet

wingspan is 35.9 feet

height is 17.3

max takeoff weight is 51,800LBS

Performance

max speed is mach 2

service ceiling is 55,000 feet

range is 1,564 nautical miles

powerplant is two Eurojet EJ200 afterburning turbofans 20,233 pounds thrust each with afterburner

Key systems and armament

radar CAPTOR active/mechanical radar systems

sensors PIRATE passive infrared search and track (IRST) system

Air to Air missiles same missiles as the other 2 jets

internally mounted 27mm Mauser BK-27 rotary cannon

MAP

400x400 Nautical miles

the map should have, Islands 1, 2, 3,

Skye:steep mountains, sea lochs, and coastal cliffs.

Capri:A compact, rocky island known for steep limestone crags and blue grottoes.

**Samos:featuring high pine-covered hills and pebble beaches

Each team gets 1 island, with 2 runways, and the last island (3rd) will have both the enemies and the users teams runway (just 1 runway for the 3rd island)and will be fully opposite from across the island (3rd), and there will be a mountain in between the runways that covers up from sight, and the user (including NPCS) needs to fly around or above in consideration of the service ceiling it to see the NPCS, (TERRAIN BLOCKS RADAR, IF THE MATH IS TOO HARD BASE IT OFF OF REAL LIFE METRICS)

the game should look like a very realistic roblox game, but more towards DCS flying sim. (uploaded pictures are reference)

(THE CAN BE CUSTOMIZABLE OF 3RD PERSON AND 1ST, WITH THE 3RD CAMERA BEING ABLE TO LOOK ALL AROUND THE VEHICLE)

GAME STUFF:

Game Mode 1: Free Flight A pure sandbox mode. The user selects any of the three authorized jets and spawns at one of their team's runways. There are no enemy engagements, allowing the user to practice flight physics, landing, and testing afterburner fuel limits across the map.

Game Mode 2: Escalating Waves Combat EngineThe strict ten-wave progressive combat challenge. Enemy NPC jets spawn with dynamically scaled difficulty and weapon behaviors. Waves 1 to 3 feature three basic jets. Waves 4 to 6 feature six tactical jets. Waves 7 to 9 feature six aggressive jets using mid-range missiles. Wave 10 features nine elite jets utilizing advanced multi-ship targeting and high-G maneuvers.

The third game mode is a customizable 1v1 duel against a single AI opponent. The user can select their own jet, the enemy's jet from the three allowed aircraft, and set the AI difficulty to easy, medium, hard, or extreme. Here is how the four difficulty settings alter the behavior of the AI pilot in the code:

Easy DifficultyThe AI flies conservatively and rarely uses the afterburner. It struggles to maintain an optimal turning rate, flies mostly in predictable straight lines, and only fires weapons when it has a perfect, sustained lock. It does not use terrain masking to hide from your radar.

Medium DifficultyThe AI begins to utilize basic defensive maneuvers like breaking away when it detects an incoming missile lock. It manages its speed to stay closer to its jet's optimal cornering velocity and uses the afterburner occasionally to regain energy during climbs.

Hard DifficultyThe AI actively attempts to get behind your jet using advanced air combat maneuvers like high yo-yos and scissors. It deploys flares and chaff defensively, aggressively uses the afterburner, and will intentionally dive behind the mountain ranges of Samos or Skye to break your radar line-of-sight.

Extreme DifficultyThe AI operates at the absolute physical limits of the aircraft. It executes flawless high-G turns just below the structural failure limit, switches instantly between guns and AIM-9X missiles based on range, and uses advanced tracking geometry to intercept you. It will aggressively exploit any mistakes you make in fuel management or G-force recovery.

ARTIFICIAL INTELLIGENCE

1. The Behavior Loop (Decision States)The AI continuously switches between four main behavioral states based on the tactical situation:

Patrol/Search: If the AI does not know where you are, it flies a scanning pattern between the islands. It uses its radar to sweep the sky, respecting the terrain masking rules. If you are hiding behind the mountain on Island 3, the AI remains in this state until it gains a clear line-of-sight ray-cast.

Intercept: Once the radar detects your jet, the AI calculates an interception vector. Instead of flying directly toward where you are right now, it calculates where you will be in the future based on your current Mach speed and heading, flying a lead-pursuit course to cut you off.

Engage (Dogfight): When within weapon range, the AI actively tries to maneuver into your six o'clock position (behind you). It calculates the optimal cornering speed for its specific jet model (e.g., managing throttle precisely to keep an F/A-18 at its best turning rate) and uses basic or advanced flight maneuvers to out-turn you.

Defensive: If the AI detects an incoming missile warning or notices you are positioned directly behind it, it immediately breaks into a hard, high-G turn perpendicular to the missile's path (beaming) and rhythmically drops flares or chaff.

2. How Difficulty Modifies the AI MathThe difficulty setting chosen by the user (or determined by the current wave in Wave Mode) acts as a modifier on the AI's reaction times, piloting precision, and tactical choices:

Easy AI: The decision loop updates very slowly (e.g., checking surroundings only once every two

seconds). It pulls gentle turns, keeping G-forces low, and never uses the afterburner. It flies in predictable arcs, making it an easy target for your AIM-9X missiles. Medium AI: The loop updates faster. It monitors its speed and uses the afterburner to recover energy after climbing over hills or mountains. It will pull up to 6Gs to try and shake you off its tail. Hard AI: The loop updates almost instantly. The AI aggressively monitors its target's position. It intentionally utilizes the vertical plane (looping upward or diving down) to trade altitude for speed. If you fire a missile, it immediately rolls, pulls a hard break turn, and cuts its throttle to reduce its heat signature against your infrared missiles. Extreme AI: The loop updates every single frame. The AI knows the exact flight envelope limits of its chosen jet. It will pull tight turns up to its maximum G-limit (11G for the F-15EX) to force a snapshot gun solution with its rotary cannon. It constantly monitors your orientation; if it sees your nose pointing away, it immediately engages full afterburner to close the distance.

3. Terrain and Environment Awareness The AI reads the heightmap coordinates of the 400x400 NM map. It evaluates the mountain peaks of Skye, Capri, and Samos. On higher difficulties, if the AI is defensive and low on fuel or options, it will intentionally dive into the sea lochs or behind the massive mountain dividing Island 3. It uses the terrain masking logic to deliberately break your radar lock, forcing you to either give up the chase or follow it down into the risky, low-altitude mountain passes where one wrong high-G turn could cause a crash or a G-LOC blackout.

GAME UI

1. Main Menu & Configuration UI This is a clean, modern interface where you set up your flight parameters before entering the 400x400 NM map. Aircraft Selector Screen: A 3D hangar showcase displaying the three allowed jets (F-15EX, F/A-18, Eurofighter). Hovering over a jet updates a side panel showing its exact real-world specs (Max Mach speed, weight, service ceiling, and weapon payload capacity). Game Mode Selector: A simple grid showing the three available options: Free Flight, Wave Mode (1-10), and 1v1 Custom Duel. 1v1 Customizer Panel: If Mode 3 is selected, a sub-menu opens allowing you to pick your opponent's jet and toggle a slider for the AI difficulty (Easy, Medium, Hard, Extreme). Open sidebar New thread Manage AI Mode shared public links See your Search history Personalization Settings New thread Search threads Close sidebar Easily pick up where you left off Follow up faster, revisit your searches, and more when you save your history Update settings (function() { window.sn = { _setImageSrc: function(b, c) { var a = document.getElementById(`\${b}`); a !== null && (a.src = c, a.setAttribute("data-deferred", "2"), google.rll(a, !0, () => { a.setAttribute("data-imgprocessed", "true") }) }); } }).call(this); AI Mode Conversation: if i were making a game and i had a jet, what would the game UI be You said: if i were making a game and i had a jet, what would the game UI be Copied Copy Edit if i were making a game and i had a jet, what would the game UI be AI Mode reply for if i were making a game and i had a jet, what would the game UI be A jet game user interface (UI) should use a transparent green or amber Heads-Up Display (HUD) overlaid directly on the screen, balancing realistic flight data with fast-paced arcade action. GitHub +1 Core HUD Elements (Non-Diegetic & Spatial) Pitch Ladder: Central horizontal lines that move up

and down to show if your nose is pointing at the sky or the ground. Velocity Vector / Flight Path Marker: A small circle with little wings showing exactly where the jet is physically heading. Compass Tape: A scrolling directional strip at the very top of the screen showing your heading (e.g., 270, W). Speed and Altitude Tapes: Vertical numerical scales sliding up and down on the left (airspeed or Mach number) and right (altitude) sides of the screen. G-Force Meter: A warning indicator showing high stress during sharp turns.

YouTube·Vazgriz +1 Combat and Weapon UI (Meta Elements) Target Reticle / Boresight: A box or circle in the center locking onto enemy aircraft when in range. Radar Mini-map: A circular radar sweep at a bottom corner showing nearby friendly and hostile blips. Weapon Loadout Status: Text or icons on a bottom corner displaying active missiles, gun ammo count, and counter-measures (flares/chaff) remaining. Throttle Gauge: A vertical meter or percentage bar indicating engine power and afterburner status.

YouTube·Vazgriz +1 Would you like to focus the UI design more on an arcade style (like Ace Combat) or a hardcore simulator (like DCS World)? Copied to clipboard Failed to copy to clipboard. Try again later. Copied Failed to copy Copy Share public link This public link is valid for 7 days and shares a thread, including any personal information you added. This link or copies made by others cannot be deleted. If you share with third parties, their policies apply. Copy link Can't copy the link right now. Try again later. Facebook Gmail X Reddit WhatsApp Share Good response Bad response More Export to Docs Draft in Gmail About this response Saved

time Clear Helpful Comprehensive Other Incorrect Inappropriate Not working Unhelpful Other A copy of this chat will be included. A copy of this chat and the content you shared will be included. A copy of this chat and the content you shared will be included. Submit Thanks for letting us know Google may use account and system data to understand your feedback and improve our services, subject to our Privacy Policy and Terms of Service. For legal issues, make a legal removal request. Shared 0 files Reddit·hbisi81 Jet Fighter Game Creator Demo WIP- URPT The flight interface displays essential flight data including air speed, Angle of Attack (AOA), and G-force loads on the left side... 1:06 YouTube·Vazgriz Flight Simulator in Unity3D: HUD (Part 2) Flight Simulator in Unity3D: HUD (Part 2) in the last video we learned and implemented basic flight physics. however there are sev... 11m Unity Discussions Jet Game - Help Required.... - Getting Started In the game you get to select between Fighter Jets and you carry out your missions with that jet. You get to shoot missiles and is my game have it make war thunder. The only Vehicles it should make is F-15 EX II, F-18 Super Hornet, and Eurofighter, NO OTHER VEHICLES, NO TANKS, NO BOATS, NO OTHER JETS EXCEPT THE ONE I SAID.

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Crew is 2

Length is 63.8 Feet

Wingspan is 42.8

Height is 18.5

Maximum Takeoff Weight is 81,000LBS

Powerplant is Two General Electric F110-GE-129 Turbofans each providing up to 29,500LBS of thrust with afterburners

Combat Range (how far it can fly): 687 Nautical Miles

Service Ceiling is 60,000

G-Limits is 11 (Pilot passes out completely at 10.5G, the screen stays black for 10s, first signs of losing conciseness, at 4g the screen starts to get grey, and 8-10Gs is monochrome. -2 G is 50% red, -5G is Only red

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Length is 60.3 feet

wingspan is 44.9 feet

height is 16 feet

crew is 2

Performance

max speed is mach 1.8

service ceiling is 50,000ft

combat range is 1,275 nautical miles

Engines and weight

Two General Electric F414-GE-400 afterburning turbofan engines

Thrust 22,000LBS per engine With afterburner

max takeoff weight is 66,000LBS

Armament

Cannon is Internal M61A2 Vulcan 20mm rotary cannon

External weapon stations (hardpoints) is 11

missiles and bombs supports AIM-120D and AIM-9X

Eurofighter Typhoon

Crew is 1 seat

length is 52.4 feet

wingspan is 35.9 feet

height is 17.3

max takeoff weight is 51,800LBS

Performance

max speed is mach 2

service ceiling is 55,000 feet

range is 1,564 nautical miles

powerplant is two Eurojet EJ200 afterburning turbofans 20,233 pounds thrust each with afterburner

Key systems and armament

radar CAPTOR active/mechanical radar systems

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the map should have, Islands 1, 2, 3,

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Each team gets 1 island, with 2 runways, and the last island (3rd) will have both the enemies and the users teams runway (just 1 runway for the 3rd island)and will be fully opposite from across the island (3rd), and there will be a mountain in between the runways that covers up from sight, and the user (including NPCS) needs to fly around or above in consideration of the service ceiling it to see the NPCS, (TERRAIN BLOCKS RADAR, IF THE MATH IS TOO HARD BASE IT OFF OF REAL LIFE METRICS)

the game should look like a very realistic roblox game, but more towards DCS flying sim. (uploaded pictures are reference)

(THE CAN BE CUSTUMIZABLE OF 3RD PERSON AND 1ST, WITH THE 3RD CAMERA BEING ABLE TO LOOK ALL AROUND THE VEHICLE)Here is a highly detailed, architecture-focused master prompt designed to get Claude to generate the core code for your flight simulator. Since Claude works best when given structured system blueprints, this prompt explicitly defines the scope, strict exclusions, aircraft mathematical physics, and math rules for the radar-blocking terrain. Copy and paste everything below into Claude to begin the codebase generation. Master Developer Blueprint: Hardcore Flight Simulator Project

□ Role & Project ScopeYou are an elite, specialized Game Developer and Flight Dynamics Engineer. Your goal is to generate a massive, fully modular, production-ready flight simulator codebase spanning 30,000 to 60,000 lines of code (structured step-by-step across multiple files or sequential prompts as needed). The game combines the visual presentation of a hyper-realistic, high-fidelity Roblox flight game with the uncompromising flight mechanics, physics, and avionics of DCS World. □ ABSOLUTE HARD CONSTRAINTS (STRICT EXCLUSIONS)ALLOWED VEHICLES: Only three aircraft are permitted in the entire codebase: F-15EX Eagle II, F/A-18E/F Super Hornet, and the Eurofighter Typhoon. STRICT BAN: NO other jets, NO propeller planes, NO tanks, NO ground units, NO ships, and NO boats are allowed in the logic, enum systems, or object models. □ SECTION 1: THE MASSIVE WORLD MAP (400x400 NM)1.1 Map Dimensions and StructureScale: A coordinate space representing exactly 400 x 400 Nautical Miles (NM) ($(1 \text{ NM} \approx 1852 \text{ meters})$). Environment: The world features procedural or highly detailed coordinate heightmaps consisting of scattered trees, hills, and steep mountains. Absolutely no grass rendering or grass physics surfaces. Islands: The coordinates house three specific, hyper-detailed island archetypes:Skye: Steep, razor-sharp mountains, dramatic sea lochs, and jagged coastal cliffs.Capri: A compact, sheer rocky island defined by steep limestone crags and deep sea-level blue grottoes.Samos: High, pine-tree-covered rolling hills and smooth pebble beaches. 1.2

Base and Airfield Layout There are exactly 6 operational airports/runways distributed evenly across the map: Team A (User Team): Controls Island 1 (Skye) which houses 2 full runways. Team B (Enemy Team): Controls Island 2 (Capri) which houses 2 full runways. The Contested Island (Island 3 - Samos): Houses exactly 2 runways total—one assigned to Team A and one assigned to Team B. The Samos Inversion Rule: On Island 3, the two runways must be placed on completely opposite sides of the island. A massive, towering mountain range must sit precisely between them, completely breaking physical line-of-sight.

1.3 Radar Line-of-Sight & Terrain Masking Math

Implement a continuous ray-casting vector check to calculate whether an aircraft's radar can detect a target behind mountains. Use this exact metric:

$$(\text{LOS}) = (\text{Ray}(P_{\text{Radar}}, P_{\text{Target}}))$$

If the intersection test returns true against the terrain heightmap, the target is completely masked. It must disappear from the aircraft's Active Electronically Scanned Array (AESA) or mechanical radar display. Pilots must fly above or around the mountain or respect the aircraft's service ceiling to reacquire the target track.

SECTION 2: THE JETS - AUTHENTIC TECHNICAL SPECIFICATIONS

Implement a rigid data structure using strict class definitions or struct objects for the three allowed jets, configuring their exact real-world envelopes:

1. F-15EX Eagle II Specs

Crew: 2 (Pilot + WSO) Physical Dimensions: Length: 63.8 ft | Wingspan: 42.8 ft | Height: 18.5 ft Weights & Propulsion: Max Takeoff Weight: 81,000 lbs. Two General Electric F110-GE-129 Turbofans delivering 29,500 lbs of thrust each with full afterburner. Performance: Max Speed: Mach 2.5. Combat Range: 687 NM. Service Ceiling: 60,000 ft. Armament: 23 weapon hardpoints supporting up to 29,000 lbs of payload. Internal M61A1 20mm rotary cannon. Supported ordinance loadout: AIM-120D AMRAAM and AIM-9X Sidewinder (up to 12 air-to-air missiles total). Avionics & Systems: AN/APG-82(V)1 AESA Radar, Eagle Passive Active Warning Survivability System (EPAWSS), and an all-digital Fly-By-Wire (FBW) system.

2. F/A-18E/F Super Hornet Specs

Crew: 2 (Pilot + WSO) Physical Dimensions: Length: 60.3 ft | Wingspan: 44.9 ft | Height: 16.0 ft Weights & Propulsion: Max Takeoff Weight: 66,000 lbs. Two General Electric F414-GE-400 afterburning turbofans delivering 22,000 lbs of thrust each with full afterburner. Performance: Max Speed: Mach 1.8. Combat Range: 1,275 NM. Service Ceiling: 50,000 ft. Armament: 11 weapon hardpoints. Internal M61A2 Vulcan 20mm rotary cannon. Supported ordinance loadout: AIM-120D AMRAAM and AIM-9X Sidewinder. Avionics & Systems: High-definition tracking radar, digital flight control systems.

3. Eurofighter Typhoon Specs

Crew: 1 Physical Dimensions: Length: 52.4 ft | Wingspan: 35.9 ft | Height: 17.3 ft Weights & Propulsion: Max Takeoff Weight: 51,800 lbs. Two Eurojet EJ200 afterburning turbofans delivering 20,233 lbs of thrust each with full afterburner. Performance: Max Speed: Mach 2.0. Combat Range: 1,564 NM. Service Ceiling: 55,000 ft. Armament: Internal Mauser BK-27 27mm rotary cannon. Supported ordinance loadout: AIM-120D AMRAAM and AIM-9X Sidewinder. Avionics & Systems: CAPTOR active/mechanical tracking radar, PIRATE Passive Infrared Search and Track (IRST) sensor suite.

SECTION 3: ADVANCED DYNAMICS & HUMAN PHYSIOLOGY SYSTEMS

3.1 Fuel Consumption Lifecycle Loop

Engines consume standard fuel reserves continuously based on throttle position. Engaging the Afterburner grants immediate maximum thrust multipliers, allowing the aircraft to punch through the sound barrier up

to its maximum Mach limit. However, it applies a massive 10x fuel depletion coefficient rate.

3.2 Human G-Force Tolerance Simulation Loop

Write a loop that evaluates structural centrifugal force loads against human physiology (G_z). Apply these screen rendering overlay effects step-by-step based on the current G-load:

Positive G-Loads: $+4.0G$ to $+7.9G$: Begin a linear desaturation screen effect. The rendering window fades gradually from full color into a grey-out state. $+8.0G$ to $+10.4G$: Enter full tunnel vision. The screen turns completely monochrome (black and white). $+10.5G$: Immediate G-LOC (G-induced Loss of Consciousness). Force a 10-second blackout window. The screen stays pitch black, and all control inputs are completely frozen or ignored until the timer expires.

Negative G-Loads: $-2.0G$ to $-4.9G$: Apply a 50% redout vignette filter over the screen. $-5.0G$: Crimson redout. The screen becomes fully saturated with deep 100% solid red, heavily obscuring vision.

SECTION 4: CORE GAME MODES & AI ENGINE

Implement two fully fleshed-out game mode directors:

4.1 Game Mode 1: Free Flight

A pure, uninhibited sandbox loop. The user selects any of the three authorized jets and spawns at one of Team A's runways, with full freedom to fly around the 400x400 NM map without combat encounters.

4.2 Game Mode 2: Escalating Waves Combat Engine

A strict 10-Wave progressive combat challenge loop. The enemy AI team must spawn with dynamically scaled difficulty, tactical behavior, and weapon selection thresholds:

Waves 1 to 3: Exactly 3 enemy NPC jets spawn. The AI uses basic flight paths, poor evasive maneuvering, and relies mostly on guns or loose missile locks.

Waves 4 to 6: Exactly 6 enemy NPC jets spawn. The AI begins using proper defensive maneuvers, breaks lock using terrain masking, and actively deploys countermeasures.

Waves 7 to 9: Exactly 6 enemy NPC jets spawn. The AI gains high aggressive tuning, utilizes afterburners intelligently, and shoots mid-range AIM-120D missiles with advanced interception geometry.

Wave 10 (The Finale): Exactly 9 elite enemy NPC jets spawn. The AI exhibits perfect tactical teamwork, applies maximum pressure using advanced multi-ship targeting, and makes aggressive use of high-G turns to win dogfights.

SECTION 5: CAMERA SYSTEMS & ARCHITECTURE

1st Person Cockpit View:

Render a fully locked internal viewpoint with a functional HUD overlay, displaying moving pitch ladders, airspeed/altitude tapes, and weapon status.

3rd Person Orbit View:

Implement an external chase camera configuration that allows full 360-degree free mouse look around the aircraft's model geometry without breaking the orientation of the vehicle's vector path.

Step-by-Step Codebase Generation Plan

Now, generate this simulator architecture layer by layer.

Module 1:

Begin by writing the core Math, Vector System, and 400x400 NM Terrain Map Generation Code (including Island locations, runways, and the Terrain/Radar Masking Interception algorithms). Let me review it, and then we will immediately proceed to Module 2: Flight Physics, Afterburner Fuel Dynamics, and the G-LOC Physiology Loop.

Show me the structural setup for Module 1 now. Would you like Claude to write this script in a specific language or engine format (such as Luau/Roblox Luau, TypeScript with Three.js, or Python/Pygame for the underlying math logic)?

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 □ Positive G-Forces (Pulling Up / Tight Turns) When the pilot pulls the flight stick back, blood drains from their head to their boots. The screen changes dynamically to show them losing consciousness: 4.0G to 7.9G (The Grey-Out): The camera's color begins to drain. The edges of the screen fade into a blurry, desaturated gray as the pilot starts losing peripheral vision. 8.0G to 10.4G (The Monochrome Tunnel): The screen turns completely black and white. The player is experiencing severe tunnel vision and is on the verge of passing out. 10.5G or Higher (G-LOC Blackout): The pilot completely passes out (G-Induced Loss of Consciousness). The screen goes pitch black for exactly 10 seconds. During this time, the controls completely freeze—if the jet was heading toward a mountain when the player blacked out, it will likely crash before they wake up.
 □ Negative G-Forces (Pushing Down) When the pilot pushes the flight stick forward, blood rushes intensely into their head and eyes, causing a dangerous condition called a Redout: -2.0G to -4.9G (Partial Redout): A crimson vignette creep into the borders of the screen, creating a 50% solid red tint over the player's view. -5.0G or Lower (Full Redout): The screen turns 100% solid, saturated red. The player's vision is completely blinded by blood pressure in the optic nerves, making it impossible to see the HUD or the terrain.
 □ How the Code Calculates This Behind the scenes, Claude will create a loop that runs every single frame. It takes the jet's current velocity vector and tracks how quickly that vector is changing. It calculates the current $[G_z]$ acceleration. It plugs that number into a graphics shader filter. If $(G \geq 10.5)$, it starts a Timer = 10.0 countdown, sets the screen to #000000, and disables keyboard/joystick inputs until the timer hits 0. Would you like to adjust these thresholds to make the game a bit more forgiving, or should we keep the hardcore realistic penalty exactly as it is? Copied to clipboard Failed to copy to clipboard. Try again later. Copied Failed to copy Copy Share public link This public link is valid for 7 days and shares a thread, including any personal information you added. This link or

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Copied Copy Edit in the game i also want to add like 3 game modes, AI Mode reply for in the game i also want to add like 3 game modes, We can expand the game mode manager module to handle exactly three distinct modes. Since we already have free flight and the ten-wave combat progression, we just need to decide on the third mode to complete the game loop. Here are the three game modes structured for the architecture prompt:

Game Mode 1: Free Flight A pure sandbox mode. The user selects any of the three authorized jets and spawns at one of their team's runways. There are no enemy engagements, allowing the user to practice flight physics, landing, and testing afterburner fuel limits across the map.

Game Mode 2: Escalating Waves Combat Engine The strict ten-wave progressive combat challenge. Enemy NPC jets spawn with dynamically scaled difficulty and weapon behaviors. Waves 1 to 3 feature three basic jets. Waves 4 to 6 feature six tactical jets. Waves 7 to 9 feature six aggressive jets using mid-range missiles. Wave 10 features nine elite jets utilizing advanced multi-ship targeting and high-G maneuvers.

Game Mode 3: [Awaiting Your Choice] To fill the third slot, we can add one of these modes to match the War Thunder style:

- Air Airfield Assault: A team-vs-team match where the user and two friendly AI jets must defend their islands while trying to eliminate the three enemy runways across the map.
- 1v1 Guns-Only Dogfight: A pure test of piloting skill where the user spawns mid-air directly facing a single elite enemy NPC jet with no missiles allowed, relying entirely on the internal rotary cannons.
- Time Trial Navigation: A non-combat mode where checkpoints are scattered tightly through the mountain crags of Skye and Capri, forcing the user to fly at high Mach speeds close to the ground without crashing or blacking out from G-forces.

Which option would you like to use for the third game mode, or do you have a different mode in mind that you want to add? Copied to clipboard Failed to copy to clipboard. Try again later. Copied Failed to copy Copy Share public link This public link is valid for 7 days and shares a thread, including any personal information you added. This link or copies made by others cannot be deleted. If you share with third parties, their policies apply. Copy link Can't copy the link right now. Try again later. Facebook Gmail X Reddit WhatsApp Share Good response Bad response More Export to Docs Draft in Gmail About this response Saved

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spam, while the Eurofighter relies more on close-in IRST tracking and dogfighting)?

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timeClearHelpfulComprehensiveOtherIncorrectInappropriateNot workingUnhelpfulOtherA copy of this chat will be included.A copy of this chat and the content you shared will be included.A copy of this chat and the content you shared will be included.SubmitThanks for letting us knowGoogle may use account and system data to understand your feedback and improve our services, subject to our Privacy Policy and Terms of Service. For legal issues, make a legal removal request.Shared0 filesShow lessShow allgoogle.xsrf = google.xsrf || {};google.xsrf = {...google.xsrf,...{"ahxKmf":"AF5tSO56wHx2pyjjGWWfRjyMKg9cgChD9g:1790380322431"}};google.xsrf = google.xsrf || {};google.xsrf = {...google.xsrf,...{"sYMMbb":"AF5tSO4w-vQBihHD-Fo3nYAPtsODgAktOw:1790380322431"}};CopiedCopyEditnow explain how the ui will work and stiffAI Mode reply for now explain how the ui will work and stiffThe UI operates across two distinct layers: the Menu Interface for selecting configurations and the In-Game Display (HUD & Camera) designed for high-speed jet combat.

1. Main Menu & Configuration UIThis is a clean, modern interface where you set up your flight parameters before entering the 400x400 NM map. Aircraft Selector Screen: A 3D hangar showcase displaying the three allowed jets (F-15EX, F/A-18, Eurofighter). Hovering over a jet updates a side panel showing its exact real-world specs (Max Mach speed, weight, service ceiling, and weapon payload capacity). Game Mode Selector: A simple grid showing the three available options: Free Flight, Wave Mode (1-10), and 1v1 Custom Duel. 1v1 Customizer Panel: If Mode 3 is selected, a sub-menu opens allowing you to pick your opponent's jet and toggle a slider for the AI difficulty (Easy, Medium, Hard, Extreme).
2. In-Game Combat HUD (Heads-Up Display)The in-game UI uses a bright, transparent green overlay aligned directly with your 1st-person and 3rd-person perspectives. It dynamically reads your flight physics data every frame. The Flight Tapes: On the left side of the screen, a vertical scrolling tape displays your current airspeed and active Mach number. On the right side, an identical tape shows your altitude in feet. Pitch Ladder & Flight Path Marker: Dead center on the screen, horizontal ladder lines tilt and shift to reflect your jet's pitch and roll relative to the horizon. A small circular icon with wings (the velocity vector) projects exactly where your physical momentum is carrying the jet. Compass Tape: A scrolling directional bar tracks across the absolute top of the screen, showing your current heading from 0 to 360 degrees. System Metrics (Bottom Left Corner): Displays critical hardware readouts:Throttle / Afterburner: A vertical percentage bar. Pushing past 100% highlights the bar in orange to show the afterburner is active.Fuel Indicator: A rapidly depleting numerical gauge, blinking red when low or when the afterburner's 10x fuel drain coefficient is active. Weapons & Combat Overlay (Bottom Right Corner): Displays your active ordinance select (AIM-120D, AIM-9X, or the internal rotary gun), remaining ammunition counts, and a count of your defensive chaff and flares.

3. Screen Distortion & G-Force EffectsThe screen itself acts as a dynamic UI element that reacts directly to your pilot's physiological state under strain.Grey-Out Phase (4.0G – 7.9G): The saturation of the entire screen gradually turns down. The edges of your vision blur and darken into a soft vignette.Monochrome Tunnel (8.0G – 10.4G): Color is completely stripped from the screen, rendering the entire game in black, white, and grey to simulate absolute tunnel vision.Blackout Freeze (10.5G+): The screen immediately cuts to a solid black overlay. A 10-second countdown timer runs silently in the code while all control inputs are ignored.Redout Overlays (Negative Gs): Pushing the stick forward creates a crimson red wash across the monitor. It stays at a 50% translucent red overlay between -2.0G and -4.9G, and locks into a 100% solid, blinding red block at -5.0G or lower.

4. Radar & Target Acquisition UIA dedicated circular radar sweep sits in the lower center of your screen to track targets across the islands.Blip Indicators: Friendly jets show as green squares, while enemy AI jets show as red triangles.Boresight Target Box: Looking directly at an enemy jet overlays a structural diamond box around them. If they dive behind a mountain on Skye, Samos, or Capri, the ray-cast line-of-sight math fails, and the radar box instantly vanishes from your screen until they clear the terrain geometry.

Vehicle Allocation and Matchmaking RulesPlayer Selection Threshold: At the match configuration menu, the user must choose exactly one aircraft out of the three allowed choices to fly for their session (either the F-15EX Eagle II, the F/A-18E/F Super Hornet, or the Eurofighter Typhoon).Dynamic AI Aircraft Restriction: The game engine must instantly read the user's selected plane asset. The spawning system is strictly forbidden from spawning that same model for any enemy NPC pilots.Opposing Team Lineup: The enemy AI forces can only be populated by the remaining two aircraft models that the user left unchosen. For example, if the user picks the F-15EX Eagle II, the AI spawning loop will generate a mixed or uniform fleet consisting entirely of F/A-18E/F Super Hornets and Eurofighter Typhoons.Zero Duplication Constraint: This lock remains active for both the 10-Wave Combat Director and the 1v1 Custom Duel mode to ensure the user is always fighting against different types of engineering envelopes.

obviously there is a lot of weird text in here. You get the general idea of the game. Build it.

ONLY HTML.

#2 • 00:16 UTC

actually build it however you want, use the github. it does not have to be only html

#3 • 04:08 UTC • 1 image attached

it says installed

#4 · 04:09 UTC

try again

#5 · 04:21 UTC

heres what i want you to do, all the jet models, i want you to fix them all, theres a lot of things wrong with it. tell me when your done with that. just make the models really high poly and not like a roblox replica.

#6 · 05:17 UTC

this update is amazing, new update for you to do, update the plane physics, like make it more realistic, also in the previous update of the jet redesign, fix the f-18, the vertical stabalizers. and on the f15 make sure they are straight, after that make the physics more realistic, not simulator style. when the stabalizers are fixed, show me picture proof.

#7 · 05:36 UTC

quick bug, when done turning in any direction while flying (all jets) it jiggles a little after, like is rotates then rotates the other way, then back, like a pendulum, but imagine turning your hand over, then over, thats what it does, fix that, then it will be good.

#8 · 05:47 UTC

dude you rock, i dont know what to say. as a celebration make the autopilot like when you click "U" it autolevels, instead of that, get rid of the autolevel, and make it an autofly like tesla full self driving lol, the user can choose speeds, and where the autopilot take is. i dont want this to be like something huge, just a small little update.

#9 · 06:05 UTC

We have a new gamemode idea. 5v5, users team vs enemys team. there will be 3 rounds, first to 3. when all 5 jets on 1 team are destroyed, the other team gets the point, if the user dies before their allies die, they can choose to spectate whoever they want including the enemy team, and go in free cam.

to make it simple basically use the code of the current enemys and like reverse it so the users teams allies fight the enemys.

you know how to make this better. this will be the major update.

#10 · 06:15 UTC

dont do the free cam if its too laggy.

#11 · 06:15 UTC

if the free cam is not laggy then add it

#12 · 06:24 UTC

now, make a new update, jet colors, add like jet customization. for all of them. add like differnt wraps and solid colors to choose from, and the brightness.

#13 · 06:25 UTC

also, make sure if a 5v5 exceeds 5 minutes, the round ends and both teams get a point.

#14 · 06:28 UTC

when custumizing the jet, make like a whole new section of the screen just for custumizing so you dont see any of the other stuff, and make srue they click like apply and stuff

#15 · 06:44 UTC · 2 images attached

make a new wrap for the jets, this style. while your at it, fix bugs that were not done from the previous update.

#16 · 06:45 UTC

call them Black Ice\

#17 · 06:58 UTC

where you at right now? hows the black ice doing?

#18 · 07:04 UTC · 5 images attached

We have a new jet. Sukhoi SU-35S Twin engine super maneuverable air superiority and multirole fighter

crew 1 pilot

length 71.9ft

wingspan 49ft

height 19.4ft

max takeoff weight 76,059lbs

powerplant Two Saturn AL041F1s afterburning turbofans with 3d thrust vectoring
thrust 32,000 per engine with afterburner

max speed mach 2.25

max range 1,944 nautical miles

service ceiling 59,060

G-limit same as other jets with the same effects

radar N035 irbis e passive electronically scanned PESA Xband radar

targeting integrated optical laser infrared search and trackIRST sensor system

armament

gun 1x30mm GSH-30-1 internal revolving rotary cannon with 150 rounds

WEAPONS ONLY FOR THIS JET! THESE WEAPONS DO NOT GO ON THE OTHER VEHICLES IN THE GAME, THESE WEAPONS ARE DESIGNED SPECIFICALLY FOR THIS JET, NONE ELSE:

R-77M

R-74M

THIS JET IS VERY MANEUVERABLE

#19 · 07:09 UTC

also the jet has a blue thrust, not orange and red and stuff

#20 · 07:11 UTC

also make it so that the ai have random wraps and colors, the users teammates and enemies in the wave mode and 5v5 mode.

#21 · 07:22 UTC

all these changes your making, add that in the update notes when your done with all this stuff

#22 · 07:25 UTC

while my users are playing the game, update the jet so its done, then in the update log put it as complete. im talking about physics, handelingm, looks, and stuff. i wont rush you.

#23 · 07:38 UTC

when the user shoots the missiles on the SU, it says Fox-3 for some reason, it needs to say what the user is shooting.

#24 · 07:44 UTC

for the f15, f18, typhoon ,when shooting the missiles its supposed to say FOX 3 instead of AMRAMAM AWAY

and the sidewinder away is for fox 2

for russina callouts look it up and make it correct.

while you do this, fix all the other jet models, the typhoon, f18, and f15.

#25 · 07:46 UTC

for f15, its fox 3, fox 2, f18, is same thing, typhoon is is an exception is fox 3 and fox 2 because it has the missiles on it, for the su35 countinue whhat you were doing

#26 · 07:48 UTC

also for the russian missile callouts, its only supposed to say the russian word.

#27 · 07:49 UTC

the russian word is "Pusk!"

#28 · 07:54 UTC

make it so in the main menu i can change to look wherever i want to see the jet, like i can move arond the jet camera, so i dont have to see it slowlyt spin.

#29 · 07:57 UTC

reverse the drag controls, and make it more smooth, and make it so i can zoom in on the jet.

#30 · 08:02 UTC

ad a XP and level system, each kill is a certain amount of XP and money, and the more XP you get the higher level you get. the things you do in game give you xp, such as like manuver, kills, and flying distances and speed. make the animations really cool, and make the leveling up fair.

#31 · 08:10 UTC

while your at it, make the exhaust for all jets like photorealistic

#32 · 08:32 UTC · 1 image attached

for the last update of the day, new map.

This time its going to be 200x200 nautical miles, half the size of map 1. (call it sometjing, then make a name for map 2). map 2 is going to be based off the reference picture uploaded but increased to 200x200 nautical miles. while your at it, make the water to look more appealing to the eye, not moving but somethin g that looks good from 20,000ft above, and make this the map. also allow the the gameoes to be played on this map.

#33 · 08:33 UTC

also make the mountains like Really fuckking tall and moutnain like,

#34 · 08:42 UTC

push to git, stop working imidietyly.

#35 · 17:27 UTC

COUNTINUE what you stopped, do the same as yesterday. Update it, put it in patch notes, fix a few bugs. Also add 3 new cool wraps.

#36 · 17:42 UTC · 1 image attached

Fix the map, it's too spikey, widen the Mountains like the reference photo uploaded, and add like a lot of tall and huge mountains. Really big.

#37 · 17:55 UTC

Even flying and I move the camera around the jet, make it SMOOTHLY reset back to its normal position on behind the jet of the chase view, make it reset after 1.8 seconds of not moving the camera, but when the user is holding the right click to change the camera, don't reset it back, once they let go then you can reset the camera

#38 · 18:32 UTC

This new update is a graphics overhaul, make the graphics changeable, like the bloom, shadows, brightness, clouds, make it 4k, make the lighting astronomically good for this game. Be able to change the overall settings as well, also in the menu change the settings Ui

#39 · 22:25 UTC

Let's add a new game mode, brainstorm a idea for one really quick, don't add delete or change anything just think of 5 cool modes

#40 · 22:27 UTC

I like the free for all, brainstorm everything that would be needed for the free for all, think outside the box, when your done with that, build it, push it. One of my requirements I want it to have is there will be 12 total jets including the user.

#41 · 23:09 UTC

The screen is black when I start flying

#42 · 23:14 UTC

Remove the XP and Level feature, it's too much. Removal every part of it but keep the game the exact same

#43 · 23:18 UTC

and the money remove that as well

#44 · 23:19 UTC · 2 images attached

Fix all the missile racks on all the jets, they need to be aligned just like the uploaded photo. Remember, all of the jet models.

#45 · 23:25 UTC

Also when you have time, when any weapon hits the jet, for exaome a missile and the jet explodes, it lags for like 2-3 seconds then goes back to normal, lagging as in screen freezing

#46 · 23:32 UTC

Also, look online examples of missile racks for each jet, don't mess with the f15, it's fine, but for the rest of the jets it's not accurate, what I want you to do for each jet is

Go online, look for missile rack examples, then add that onto the jet.

Do that for the rest of the jets

#47 · 23:45 UTC

Last thing. I want you to make an entirely new sound system for every single audio sound. Everything needs to sound new and better and not a constant annoying fan sound. Also make every single jet model new. Take your time. Make everything more realistic in the models

Sunday, September 27, 2026

#48 · 00:22 UTC

Last thing, update the physics. And fix all bigs you can find then your done.

#49 · 00:34 UTC

Finish right now. Complete and push to git

#50 · 03:30 UTC

What would be like the biggest feature of this game that really clings people? Something we can build? Just think of 5 quick things

#51 · 03:30 UTC

I have an idea, let's make it multiplayer

#52 · 04:05 UTC

how do i host all the servers

#53 · 04:08 UTC · 1 image attached

(images only)

#54 · 04:10 UTC

just push to git why do i need to pull

#55 · 04:13 UTC · 1 image attached

(images only)

#56 · 04:15 UTC

dont i need to do the custom domain thing?

#57 · 04:16 UTC · 1 image attached

(images only)

#58 · 04:22 UTC

so how are these servers running?

#59 · 04:22 UTC

how do i see the dashboard

#60 · 04:49 UTC

Lets make a new update. Fix the jet models, some of them have like upside down features.

#61 · 04:51 UTC · 1 image attached

look at the intake area

#62 · 05:05 UTC · 5 images attached

new jet idea. build the jet, make the model perfect, including the specs, and the physics. you might need to look how it looks because i cant add more pictures.

#63 · 05:07 UTC · 1 image attached

(images only)

#64 · 05:10 UTC

also when you put it in the release notes, make it sound like super scary like the final boss is coming

#65 · 05:14 UTC

also add like a scary animation when opening the update notes, like take everything off the screen except for the creepy animation of the welcoming of the new boss

#66 · 05:16 UTC

[uploaded file] make it like creepy low brightness, dark red, and have scary music like this (uploaded attachment)

only play this audio for 7 secods then slowly fade out then it never plays again no matter how many times you open it. and make the animation on screen have fading transitions to match the music

#67 · 05:18 UTC

dont make it say the final boss, find like 2 sentences of detail of what the jet can do, like the amount of destruction it has.

#68 · 05:19 UTC

make the description of what it can do sound like your fucked if you try and fight itm,

#69 · 05:26 UTC

hows it lookin?

#70 · 05:28 UTC

make sure no audio is SAYING the text in the animation, the user reads it theriself in their head

#71 · 05:29 UTC

dont push the update, i will tell you when.

#72 · 05:30 UTC

countinue and push

#73 · 05:37 UTC

last thing, make every single jet model 1000x more realistic. then your done. make sure every single one is FULLY complete and photorealistic then you can move on. basically make it like 10x more poly

#74 · 05:49 UTC

what could be another cool feature?

#75 · 05:51 UTC · 3 images attached

add a weather widget on the top left, where you can change the weather options. rainy, snowy, clear, the clouds, etc. also make sure there is an arrow so if you click it the widget goes offscreen, then just click the arrow and its back. but this means you need to make rain, snow, clear, and cloudy.

also while your at it, make the clouds so much more, rn they are tiny dots not even scattered

#76 · 06:09 UTC

hows it lookin?

#77 · 06:10 UTC

finish and push

#78 · 06:12 UTC

how are they supposed to change it when their mouse is locked as steering?

#79 · 06:15 UTC · 1 image attached

when flying the rain is so much, make it look like water droplets on the screen like the uplaoded image abve, and not making it look like your traveling like lightspeed

#80 · 06:15 UTC

also when looking at the horizion and looking up and down the clouds appear and reappear

#81 · 06:18 UTC

hurry up.

#82 · 06:20 UTC

dont do any f8 or change the mosue, it already worjs

#83 · 06:23 UTC · 1 image attached

notice how it is poppig up in the middle? i only want it to spawn on the VERY side of the screen, oterwise its perfect, and make them 3 time ssmaller

#84 · 06:23 UTC

hurry up PLEASE

#85 · 22:38 UTC

Start doing many big fixes after you complete this task.

When shooting at an enemy, there will be a little circle showing where to shoot so you know where to shoot your main gun so it hits them, kind of like war thunder.

#86 · 23:19 UTC

new jet. This will be the first gen 5,

Dimensions & Weight

- * Length: 62 feet (18.9 meters)
- * Wingspan: 44.5 feet (13.6 meters)
- * Empty Weight: ~43,300 lbs (Heavy fighter class)

Speed & Performance

- * Top Speed: Mach 2.25 (~1,500 mph / 2,414 km/h) at high altitude
- * Supercruise Speed: Mach 1.8 (Speed it can sustain without using afterburners)
- * Climb Rate: 60,000+ feet per minute (Rockets vertically instantly)
- * Service Ceiling: 65,000 feet (Can fly higher than most 4th-gen jets)

Propulsion (The Engines)

- * Engines: 2× Pratt & Whitney F119-PW-100 turbofans
- * Total Thrust: 70,000 lbs with full afterburners
- * Thrust-Vectoring: Engine nozzles can pitch up or down by 20 degrees for impossible, tight flips and post-stall maneuvers.

Standard Weapons Payload (Internal Bays)

- * Gun: 1× 20mm M61A2 Vulcan rotary cannon (480 rounds)
- * Air-to-Air Setup:
- * 6× AIM-120 AMRAAM (Long-range radar missiles)
- * 2× AIM-9 Sidewinder (Short-range heat-seeking missiles)

I DO NOT WANT THIS JET TO BE OVERPOWERD, THE ONLY REASON IM ADDING THIS JET IS FOR THE SPEED. and acceleration. dont add like a insanely crazy radar, maybe just like the f15s tech specs, but for performance wise use what the text above says.

#87 · 23:20 UTC · 5 images attached

(images only)

#88 · 23:26 UTC

i also really want you to make this model good, make sure its photorealistic, i want it to be also the 10 million polys you did for the other jets do the same for this or make it even better. i want this to be the coolest looking one and most realistic.

#89 · 23:31 UTC

the jet looks so bad, i want you to spend like 10 minutes making it perfect because it looks like a shitty roblox addon. take all the time you need, i want it to look really good, while your at it make sure the artifact is updated and you dont need my permission to actiate the update nor push it.

#90 · 23:33 UTC

matter of fact, take 20 minutes to finish it

#91 · 23:45 UTC

now, create like a SUPER FUCKING long pdf file of everything i ever asked you to make the game. from day 1 to now.

Part 2 — What got built: every release

The release notes for every version, newest first, exactly as they appear in the game under NOTES and in CHANGELOG.md.

v4.5.1 — F-22A model rebuilt

2026-09-28

- F-22A model rebuilt from the ground up. The flat, faceted fuselage now has a sharp chine running from the nose to the tail and the wide flat deck over the intakes and wing roots. The caret intakes are built into the sides of the fuselage, with dark ducts. It also has the frameless gold canopy on a raised sill, broad trapezoidal fins canted out 28 degrees, the tail booms that carry the all-moving tailplanes, and square 2D nozzles with serrated upper and lower flaps that swing as you vector.
- Correct stance: the F-22 now stands 16.7 ft tall on its gear, with the nose leg behind the radome.
- Your own F-22 and the hangar jet are built at about 3.8 million triangles, the most detailed jet in the game.

v4.5.0 — F-22A Raptor

2026-09-28

- NEW JET: the LOCKHEED MARTIN F-22A RAPTOR, the first fifth-generation fighter in the game, and the fastest.
- Speed: supercruise at Mach 1.8 with no afterburner, Mach 2.25 flat out, a 65,000 ft ceiling, and a climb that goes straight up at around 60,000 ft a minute. It's the quickest jet in the game from Mach 0.9 to 1.6 and to 36,000 ft.
- Engines: 2x Pratt & Whitney F119-PW-100, 70,000 lbf total with afterburner. The 2D nozzles vector 20 degrees up and down (pitch only) for tight flips and post-stall moves with G-limiter override.
- Kept fair on purpose: the radar is F-15EX-class, the radar signature is normal (no stealth advantage), and it carries only 8 missiles, all in internal bays: 6 AIM-120D and 2 AIM-9X. M61A2 20 mm gun, 480 rounds. Missiles in the bays add weight but no drag.
- Dimensions: 62 ft long, 44.5 ft span, 16.7 ft tall, empty weight 43,300 lb.
- New 3D model, the most detailed in the game (your own jet and the hangar get an even denser build than the others). It has the chined diamond nose, a frameless gold-tinted bubble canopy, caret intakes, a diamond wing with a forward-swept trailing edge, big all-moving tailplanes, twin fins canted out 28 degrees, flat two-dimensional nozzles that move as you vector, and the two-tone Raptor grey with its darker patches.
- The F-22A is in every mode, including multiplayer on the official servers.

v4.4.1 — Fixes: clouds, reflections, weather panel

2026-09-28

- Fixed: a regular micro-stutter in cloudy weather. Up to 20,000 cloud puffs were re-sorted five times a second with a slow sort; they now use a fast native sort.
- Fixed: jets reflected a bright blue sky in rain, storms and overcast. Paint and canopy reflections now match the weather, so they're grey under the clouds.

- Fixed: on smaller screens the weather panel covered the FLIGHT panel. On short screens it now starts tucked away as just the arrow tab, and it's a little smaller. It remembers whether you left it open.
- Performance: when it isn't raining, the screen-droplet layer no longer redraws and re-uploads its texture every frame.

v4.4.0 — Gun lead marker

2026-09-27

- NEW: gun lead marker, like War Thunder. When an enemy is inside gun range (about 1 NM), a small circle appears ahead of them, with a dotted line from the jet, showing exactly where to shoot. Put your gun cross on the circle and your rounds meet the target: it allows for the target's motion, your own speed, the rounds slowing down and bullet drop. It turns red when you're on target. It works for the locked target, or else for the enemy nearest your gun line, with any weapon selected, in cockpit and outside views.

v4.3.2 — Smaller rain drops

2026-09-27

- Rain drops on the screen are a third of the size and land only along the very left and right edges, so the middle of the screen stays clear.

v4.3.1 — Rain on the canopy

2026-09-27

- Rain now beads on the screen like water on a window: drops land, grow and refract a blurred, flipped view of what's behind them. At speed the airflow sweeps them sideways off the edges of the screen; when you're slow they run down and slide off.
- The falling rain no longer streaks past like you're jumping to lightspeed: the streaks are short, and they thin out the faster you fly, so the drops on the canopy take over.
- Fixed: clouds popping in and out at the horizon. The far edge of the cloud field now fades out smoothly instead of cutting off, and the cloud deck reaches well past the horizon.

v4.3.0 — Weather

2026-09-27

- NEW: WEATHER. A weather panel sits at the top left while you fly: pick CLEAR, CLOUDY, OVERCAST, RAIN, STORM or SNOW, then fine-tune CLOUD COVER, RAIN / SNOW and VISIBILITY (shown in NM). The arrow tab slides the panel off the screen and back. Your choice is saved.
- New clouds: big, full cumulus built from overlapping puffs and spread across the whole theater, from a few fair-weather clouds on a clear day to a sky full of towering cumulus. They no longer look like small clumps of dots.
- Overcast, rain and snow bring a solid cloud deck. Under it the sun is hidden and the day goes grey. Fly up into it and you're in whiteout, then break out on top into sunshine over a sea of cloud.
- RAIN: a dark deck, rain streaks that stretch into lines at speed and lower visibility. STORM: heavier rain, darker cloud, thunderheads, and lightning that lights up the sky, followed by thunder. SNOW: drifting flakes in grey murk. Rain and snow fall only below the cloud base.

- Weather only changes your own screen: in multiplayer, other pilots see their own weather. The old CLOUDS option in Settings is replaced by the weather panel. CLOUD QUALITY still sets how full each cloud is.

v4.2.0 — Hero-detail jets

2026-09-27

- Every jet now has a HERO model with about 10 times the polygons: 2.5 to 4 million triangles per airframe (F-15EX 3.3M, F/A-18E/F 2.9M, Typhoon 2.5M, Su-35S 4.0M, Rafale 2.6M), up from 250,000 to 390,000.
- The whole airframe is rebuilt at the higher density: fuselage lofts, intakes and ducts, canopies and frames, wings, canards, fins and control surfaces, nozzles and petals, landing gear and every turned part. Curves are smooth all the way round and silhouettes stay clean however close the camera gets.
- You get the hero model for your own jet and in the hangar. The jets around you keep the standard model so a 12-jet fight still runs smoothly.
- Hero density follows GRAPHICS QUALITY in the settings: about 10x on HIGH and ULTRA, about 4x on MEDIUM and about 1.5x on LOW.
- Memory: the hangar now keeps only the jet on the turntable built and frees the others. A hero model is freed when nothing uses it any more.

v4.1.0 — The Rafale is coming

2026-09-27

- It's here. NEW JET: the DASSAULT RAFALE C. You don't hear it coming. Your radar warning receiver is the first thing that knows, and by then it's already too late.
- It sees you first: RBE2 AESA radar out to 90 NM, OSF passive infrared search and track out to 42 NM, and SPECTRA self-protection that jams your missiles and dumps chaff and flares on its own.
- METEOR: a ramjet air-to-air missile that stays under power all the way to you. It reaches about 92 NM from 40,000 ft (about 52 NM at 20,000 ft), more than double the AIM-120D. Inside about 34 NM there is no escape: no turn, no dive and no afterburner outruns it.
- MICA IR: an imaging heat-seeker with thrust vectoring that reaches about 27 NM, out-reaching every other heat-seeker in the game. It rides on the wingtips and pylons.
- 14 hardpoints and 20,900 lb of payload: up to 10 air-to-air missiles, plus a 30 mm Nexter 30M791 cannon with 125 rounds at 2,500 rounds a minute.
- Two Safran M88-2 engines (11,240 lbf dry and 16,860 lbf with afterburner each) push the lightest jet in the theater to Mach 1.8 and 50,000 ft. Its close-coupled canard delta holds energy in the turn, pulls +9 G and keeps flying at 30 degrees angle of attack. Combat range is 2,000 NM with tanks.
- Dimensions: 50.2 ft long, 35.8 ft span, 17.4 ft tall. Empty weight 21,700 lb, max takeoff 54,000 lb, g limits +9 / -3.2.
- New 3D model: the long pointed radome, OSF sensors ahead of the windscreen, the fixed refuelling probe curving forward on the right side, bubble canopy, big canards over the D-shaped side intakes, cropped delta wing with slats and elevons, wingtip missile rails, and the single fin with its squared SPECTRA fairing. Two close-set nozzles, twin nose wheels and French grey paint. It has its own engine sound, cockpit displays and stores page.

- Loadouts: AIR SUPERIORITY (4 Meteor, 4 MICA), COMBAT AIR PATROL (2 Meteor, 4 MICA, 3 tanks), MAX AAM (6 Meteor, 4 MICA) and DOGFIGHT (2 Meteor, 6 MICA).
- The Rafale is in every mode, including multiplayer on the official servers. Pick it in the hangar... if you'd rather be the one hunting.

v4.0.1 — Intakes the right way up

2026-09-27

- Fixed: the Su-35S and Typhoon engine intakes were raked upside down, with the bottom lip sticking out in front. Now the top edge leads and the mouth leans back underneath, as on the real jets: the Flanker's intakes are cut back underneath, and the Typhoon's upper lip (by the splitter) sits ahead of its lower lip.

v4.0.0 — Multiplayer

2026-09-27

- NEW: MULTIPLAYER. Press MULTIPLAYER ► on the main menu to fly LAST PILOT STANDING against real people. It's free-for-all only, and there is no AI on any server.
- Official servers OFFICIAL 1 to 5 (1-3 on Triad Isles, 4-5 on Frostfall Strait) are listed with live pilot counts. You can also join any server by typing its address.
- Host your own server: 'cd server && npm install && npm start' (options: --name, --map triad|frost, --port, --max). Up to 12 pilots per room.
- How a match works: between matches everyone flies around the arena with weapons on hold. When two or more pilots are in, a 15-second countdown starts, then everyone drops in on a ring facing the middle. Weapons are free after 8 seconds, the zone shrinks in stages and the storm outside takes you down, each kill rearms you with a missile of each type, gun rounds, flares and fuel, and the last jet flying wins. Results show for a few seconds, then the next match starts by itself.
- Join during a match and you spectate until the next one ([TAB] next pilot, [F] free camera). If you're shot down, you watch the rest of the match.
- Everything is shared between players: jets with their own paint jobs and loadouts, afterburner, gear and damage smoke, missiles in flight (with RWR and missile warnings), flares and chaff (they can decoy your missiles), gunfire tracers, and radar locks on your RWR.
- What the shooter sees decides a hit, and the pilot who was hit takes the damage and reports the kill, so the kill feed and scores match for everyone. Other jets are smoothed over the network so they fly smoothly.
- Joining a server on a different theater loads that theater, then joins automatically. Pausing doesn't stop an online match. The HUD shows your ping.
- Play online from the website version: browsers only allow secure (wss://) connections from a web page.

v3.1.0 — Physics update and bug fixes

2026-09-27

- PHYSICS: the wings now feel the wind on the runway too. Into a headwind you lift off sooner and roll shorter (about 20% less runway with a 20 kt headwind); with a tailwind it takes longer.
- Jet wash and wake turbulence: flying close behind another jet puts you in its trailing vortices, so you get bumps, sink and a roll kick near a vortex core. It's strongest right behind a heavy, slow,

hard-pulling jet and fades out about a kilometre back.

- Gun recoil: firing the cannon pushes back on the jet (about 16 kN for the F-15's M61), so long bursts cost some speed.
- Sideslip drag: flying sideways on the rudder now costs energy, as it does on a real jet.
- Negative-G overstress: pushing well past the negative G limit now over-stresses the airframe, and extreme negative G breaks it.
- Engines spool up and down more slowly in thin air at high altitude.
- External stores now add drag gradually through the transonic range, instead of jumping suddenly at Mach 1.
- Missile rocket motors produce a little more thrust at altitude, where there's less air pressure pushing back on the nozzle.
- Chaff clouds and flares slow down and drift with the wind.
- Fixed: cannon rounds could pass straight through a wing without hitting.
- Landing grades now use your airspeed at touchdown, not your ground speed.
- Fixed: in free-for-all, the drop-in countdown said 'weapons free in 6 s' but nothing held fire; nobody can shoot until weapons free now.
- Fixed: AI pilots low on fuel in free-for-all flew 'home', straight out of the zone and into the storm; they now fight on.
- Fixed: respawning in Free Flight with the fire key left the camera stuck on the death view, with the old gear and HUD state.
- Fixed: hit markers appeared for Su-35S missile hits; they're for gun hits only.
- Fixed: Su-35S messages named the AIM-9X and PIRATE; they now say R-74M and OLS-35.
- Fixed: the Duel record counted mutual kills as losses (now shown as draws); restarting a Duel or Waves mission left decoys in the air; restarting Waves from wave 1 didn't reset the mission clock.

v3.0.0 — New sound engine and realistic jet models

2026-09-27

- ALL-NEW SOUND ENGINE: every sound in the game has been rebuilt from scratch. It is all generated live (no recordings), so it follows exactly what your jet is doing.
- Fixed: the constant 'fan' drone is gone. It came from the old gun sound, which kept humming quietly in the background even when you were not firing.
- Jet engines: a layered engine sound (deep exhaust rumble, roar and jet hiss) that breathes with natural turbulence instead of a steady hum, plus a faint turbine whine that rises with the throttle. Each jet sounds different: F-15EX (F110), Super Hornet (F414), Typhoon (EJ200, the highest-pitched) and a deeper, heavier Su-35S (AL-41F1S).
- Where you listen from matters: behind a jet you hear the roar, in front of it the turbine whine, and in the cockpit everything is muffled through the canopy. In the outside views the sound fades with distance.
- Afterburner: a deep rumble with popping crackle, and a 'whump' when it lights.
- Airflow: wind rush that grows with speed, a canopy whistle, buffet when you pull hard or fly at high angle of attack, a rumble with the gear down, and the hiss of the cockpit air conditioning.
- Other jets that pass close by roar past with a Doppler pitch shift.

- Guns are rendered shot by shot at each gun's real rate of fire: the M61's BRRRT spins up and down, and the Typhoon's 27 mm and the Su-35S's 30 mm cannons thump.
- Also new: missile launches, explosions (delayed and muffled with distance), hits, flares and chaff, gear and mechanical clunks, the Sidewinder growl, RWR tones and warnings. A limiter keeps loud moments from distorting.
- MORE REALISTIC JET MODELS, all four jets:
- Paint: panel seams are now engraved into the skin and catch the light, with subtle paint mottling, varying gloss, and grime along the belly and toward the tail.
- Engine nozzles rebuilt: overlapping petal plates, the actuator ring and rods, and heat-stained metal (pale bronze at the root, straw, then blue toward the exit, sooty at the lip). Inside they are now dark and sooty with the afterburner flame holders visible, instead of a bright pale tube. At normal power they no longer glow in daylight; only the afterburner lights them up.
- Thinner, more realistic canopy frames.
- Su-35S: a proper interlocking splinter camouflage in three blue-greys (replacing the patchy see-through shapes), a slightly drooped radome like the real Flanker, and more raked intakes. The dorsal airbrake is gone, as on the real Su-35S; the rudders now splay outward to act as the airbrake.
- Typhoon: the PIRATE infrared sensor ahead of the windscreen, the BK-27 cannon blister in the right wing root, the refuelling probe fairing, and a deeper chin intake. Its splitter plate is now painted (it used to show as a black block).
- F/A-18E/F: the refuelling probe hump on the right side of the nose.
- F-15EX: gets the new paint, nozzles and canopy; its shape is unchanged.

v2.2.3 — Accurate missile racks

2026-09-26

- Missile racks researched from the real jets (the F-15EX is unchanged):
- F/A-18E/F Super Hornet: AIM-9X on the wingtip LAU-127 rails, a single missile hung under each outboard pylon, and an LAU-115 twin rack on each middle wing pylon with two AMRAAMs side by side on LAU-127 shoulder rails (as in the Navy's 'Murder Hornet' air-to-air loadout), plus the fuselage cheek AMRAAMs. A fuel tank on the middle station hangs from the pylon centre.
- Eurofighter Typhoon: back to one missile per wing pylon, as on the real jet (no twin racks), with the AMRAAMs semi-recessed under the fuselage.
- Su-35S: one missile per hardpoint as on the real Flanker, on chunkier Russian APU-170 / P-72 style launchers: R-74M on the wingtips and outer pylons, R-77M on the inner pylons, under the engine nacelles and in the tunnel between the engines.
- Loadouts, missile counts and handling are unchanged.

v2.2.2 — No more freeze on kills

2026-09-26

- Fixed: the game froze for a couple of seconds every time a jet was shot down. When a jet was destroyed, its paint job was copied to scorch the wreck, and each copy needlessly converted the whole livery (large texture data) to text, once for every part of the airframe. The wreck is now scorched with a lightweight copy, shared across parts: the kill that took seconds now takes about 3 thousandths of a second, so explosions play smoothly.

v2.2.1 — Aligned missile racks

2026-09-26

- Missile racks rebuilt on all four jets to match the real thing: missiles now ride in matched pairs on twin-rail racks, side by side on the shoulders of a shared pylon (like LAU-128s on an F-15 pylon), level with each other, parallel to the fuselage and with their noses lined up.
- F-15EX: both wing pylons are twin racks, four missiles per wing in two neat pairs, plus the tandem pairs along the conformal tanks.
- F/A-18E/F: the outboard wing pylon is a twin rack, level with the inboard pylon. Typhoon: the wing missiles pair up on one twin rack. Su-35S: the inner wing pair shares a twin rack and the outboard missile rides a shoulder rail at the same height.
- Fuel tanks on a rack station hang from the middle of the pylon, below the shoulder missiles. Loadouts, weapon counts and handling are unchanged, and missiles launch from their new rail positions.

v2.2.0 — XP and levels removed

2026-09-26

- Removed the pilot XP, level and money system completely: no pilot card on the main menu, no XP pop-ups or XP bar in flight, no level-up celebration, no XP section in the debrief, and its saved data is cleared. Everything else plays exactly the same.
- Free-for-all: the bounty and your placing work as before, just without XP or money attached.

v2.1.1 — Black screen fix

2026-09-26

- Fixed: the screen could go completely black as soon as you started flying on some graphics cards. A very bright sun glint off glossy paint or a canopy could overflow the HDR picture buffer, and the new bloom smeared that broken pixel across the whole screen. The picture is now cleaned before bloom, so this can't happen any more, and bloom still glows as before.
- Hardened the haze, cloud and free-for-all storm-wall shaders against the same kind of invalid values.

v2.1.0 — Free-for-all: Last Pilot Standing

2026-09-26

- NEW GAME MODE: FREE-FOR-ALL — LAST PILOT STANDING. 12 jets in the sky: you and 11 AI pilots, every jet hostile to every other jet. No teams, no wingmen, no respawns. Last jet flying wins.
- Drop-in: all twelve jets start spread around a ring over the contested island, facing inward at staggered altitudes, with a short countdown before weapons are free.
- The shrinking zone: the battle zone closes in stages toward a random final circle (40 NM down to 3 NM, then SUDDEN DEATH). Outside it the storm tears your jet apart, slowly at first and faster every stage; the screen glows purple and a countdown tells you how long you have. Back inside, the airframe slowly recovers. You see the storm as a towering wall of light in the sky, and on the minimap and the map [M] as a solid ring (the zone now) and a dashed ring (where it goes next).
- Scavenging: every kill puts a missile of each type back on your rails and refills some gun rounds, flares, chaff and fuel. No ground crews in a free-for-all.

- **Bounty:** the pilot with the most kills (2 or more) carries a gold bounty. Their position is revealed to everyone, with a gold marker on your HUD and minimap, and every AI hunts them. Claiming it pays +150 XP and \$600. If you carry it, everyone is coming for you.
- **Smarter free-for-all AI:** pilots pick off damaged jets, third-party fights that are already going on, hunt the bounty and reposition into the next circle before it closes. In the final circles every jet is revealed on radar and nobody can hide behind the mountains, so matches end in a fight instead of a stand-off.
- **FINAL DUEL:** when two jets are left, both are fully rearmed for the showdown.
- **Shot down?** Spectate the rest of the match: watch any pilot, see everyone's kills and who has the bounty, fly a free camera, press [T] to fast-forward 4x, or quit from the pause menu.
- **Results:** your placing out of 12 with a full scoreboard (placing, kills and when each pilot went down), then XP and money: 1st pays 600 XP / \$3,000, 2nd 350 / \$1,500, 3rd 250 / \$1,000, and everyone else earns XP for each pilot they outlasted.
- **Match options on the main menu:** AI difficulty, opponent jets (all four types, or all the same as yours), match pace (Quick ~6 min, Standard ~9 min, Long ~13 min) and weapons (all, heaters + gun, or guns only). Works on both maps.
- **Logbook:** a new FREE-FOR-ALL section (matches, wins, top-3 finishes, best placing) and three new decorations: LAST PILOT STANDING, PODIUM and BOUNTY HUNTER.
- **Fixed:** Su-35S kills with the GSh-30 cannon, R-77M and R-74M never counted toward the gun-kill, long-shot and knife-fight decorations.
- **Fixed:** the theater map [M] always said 400 × 400 NM; it now names the map you are flying and its real size.

v2.0.0 — Graphics overhaul

2026-09-26

- **GRAPHICS OVERHAUL:** every part of the picture is now adjustable, from one overall preset down to individual effects, and changes apply instantly while you watch.
- **Overall quality presets:** LOW, MEDIUM, HIGH, ULTRA and 4K ULTRA. Pick one to set everything at once; change any single option and the preset shows CUSTOM.
- **4K:** RENDER RESOLUTION can be NATIVE, 1080p, 1440p or 4K. 4K renders a true 3840 × 2160 picture (downsampled to your screen if it is smaller) for razor-sharp jets and ridgelines. A resolution scale slider fine-tunes it, and the settings show exactly what is being rendered plus your live frame rate.
- **Anti-aliasing:** OFF, MSAA 2x, 4x or 8x smooths jagged edges. The main-menu hangar now gets anti-aliasing, bloom and the picture settings too.
- **Mountain lighting:** the whole theater is lit from the real sun. Mountains cast long, soft shadows across valleys, fjords and the sea (huge at dawn and dusk), and deep valleys and gorges get less sky light while peaks are fully lit. Shadowed snow picks up the blue of the sky, and the water in a mountain's shadow loses its sun glitter.
- **Cloud shadows:** every cumulus casts a soft shadow that drifts across the land and the sea below it; mountains above the cloud base stay in the sun.
- **Light scattering:** the haze glows around the sun like real sunlight through air, strongest at dawn and dusk, and backlit clouds get bright silver linings. A low sun warms the horizon around it.
- **Bloom:** very bright light (the sun, afterburners, flares, explosions, runway lights) glows softly. Adjustable from OFF to 150%.

- Shadows: OFF, LOW, MEDIUM, HIGH or ULTRA, from 1024 up to 8192-pixel shadow maps with softer edges on the higher tiers.
- Picture: BRIGHTNESS, CONTRAST and SATURATION sliders, a lens VIGNETTE and three TONE MAPPING styles (Neutral, Filmic and the soft, natural AgX), with a one-click RESET PICTURE.
- Clouds: new OVERCAST option (Clear, Scattered, Broken, Overcast), plus CLOUD QUALITY from Low to Ultra for fuller, rounder cumulus.
- Redesigned SETTINGS: a new panel slides in from the right with a see-through backdrop so you can judge graphics changes on the jet or the world behind it. It has side tabs (Graphics, Controls, Audio, Gameplay), preset cards, clear sections and a short explanation under every option, plus sliders that show their values and on/off switches. It works from the main menu and the pause menu, and it fits phone screens too.
- Fixed: the loading screen always said 'GENERATING THE 400 × 400 NM THEATER'; it now names the map you're loading and its size.

v1.9.2 — Chase camera auto-recenter

2026-09-26

- Chase camera auto-recenter: when you look around your jet in the chase view, the camera now glides smoothly back to its normal position behind the jet after 1.8 seconds without moving it. It eases in gently, sweeps home the short way round (even after a full orbit) and settles softly, with no snapping.
- It never pulls the view away while you're still looking: as long as you hold the right mouse button (or the middle button, or keep your finger on the touch look area), the camera stays exactly where you put it. The 1.8-second timer only starts once you let go.
- The recenter only applies to your own jet in flight. The cockpit view, the spectator camera and replays keep the view where you leave it.

v1.9.1 — Giant mountains

2026-09-26

- Frostfall Strait mountains rebuilt: no more spiky knife-edge peaks. The mountains are now broad and massive, like Mount Fuji: huge snow-capped cones 20 to 40 miles across, with long sweeping concave flanks rising to a small summit crater, and gullies and ribs running down their sides.
- Lots more big mountains: every landmass is covered in these giant cones, many of them rising over 20,000 ft and the biggest above 26,000 ft, standing on wide rounded ranges instead of jagged ridges. Hvitøy's dividing wall is a broad mountain wall now too, and the rocky islets are small snow cones.
- Real snow line: the lower slopes of the mountains show bare russet volcanic rock and scree streaking up the gullies, with the snow cap above, like the reference photo, so you can see how tall they really are from the air.
- All six Frostfall airfields keep clear approach valleys through the new mountains.

v1.9.0 — Frostfall Strait, new water, 3 new wraps

2026-09-26

- NEW MAP: FROSTFALL STRAIT, a frozen arctic archipelago of 200 × 200 NM (half the size of the first map), and the new default theater. Five big snow-covered landmasses (NORDLAND and SØRVIK for BLUE, ØSTMARK and KRAGFJELL for RED) with the contested island of HVITØY in the

middle of the strait, plus scattered rocky islets (SKJÆR).

- Really tall mountains: jagged, ridged ranges climb straight out of the sea to around 26,900 ft, with sharp knife-edge ridges, bare rock on the steep faces, snowfields and glaciers everywhere else, and ice cliffs at the shoreline. Deep glacial valleys lead into every runway so approaches stay clear, and a wall of peaks splits Hvitøy between its BLUE and RED airfields.
- Pack ice: the sea around the coasts breaks up into white ice floes separated by dark leads of open water, just like the real Arctic.
- Six new airfields: NORDHAVN AB, ISVIK AB and HVITØY WEST AB (BLUE); KRAGEN AB, SVALBRU AB and HVITØY EAST AB (RED), each with its own TACAN channel. The cockpit kneeboard lists the airfields of whichever map you're flying.
- The original map is now called TRIAD ISLES (Skye, Capri and Samos, 400 × 400 NM).
- Theater picker: choose FROSTFALL STRAIT or TRIAD ISLES on the THEATER card on the main menu. Your choice is saved; the game reloads the world when you switch.
- Every game mode works on both maps: Free Flight from any friendly base, Waves, 1v1 Duel (head-on or from the runways on either side of the contested island's mountain wall) and 5v5 Team Battle. On Frostfall Strait the fights start around 29,500 ft, above the peaks, and the AI terrain-masks through the valleys.
- New water on both maps: a calm, still sea designed to look good from 20,000 ft up. It is deep navy far out, blending to rich blue and bright turquoise over shallow water near the coasts, following the real sea floor, with soft broad variations and a gentle sheen instead of busy moving waves. Frostfall has colder, darker arctic blues.
- 3 NEW WRAPS: INFERNO (charred black plates split by glowing molten veins that get hotter toward the tail), AURORA (night-sky navy with rippling green-to-violet northern-light curtains and stars) and GALAXY (deep space with purple and blue nebula clouds, dark dust lanes and glowing stars). All three glow softly in the dark. Pick them in CUSTOMIZE; you can recolour them with the base and pattern colours like any other wrap.
- Fixed: the WHAT'S NEW window could pop up over the top of a flight that had already started.
- Fixed: on the theater map (M), every island name after the first was drawn in tiny text instead of the large label.
- Fixed: mode descriptions, mission briefings and the free-flight base list no longer name the old islands when you're flying the other map.

v1.8.0 — Pilot XP & levels, realistic afterburners

2026-09-26

- NEW: Pilot XP, levels and money. Everything you do in the air earns XP; XP raises your level (1 to 100) and your rank, from CADET to GENERAL OF THE AIR FORCE. Your pilot card with level badge, rank, XP bar and money is at the top of the main menu, and it counts up your gains when you come back from a sortie.
- Kills pay XP and money, scaled by the enemy difficulty (Easy 60% up to Extreme 140%): 100 XP and \$400 per kill at Hard, plus GUN KILL (+50 XP, \$200), LONG SHOT past 20 NM (+40, \$150), DOUBLE KILL within 12 s (+50, \$250) and FIRST BLOOD (+25).
- Combat and flying XP: defeating a missile fired at you (+25 XP, \$100); manoeuvres like HIGH-G TURN (7 G for 3 s), LOOP, AILERON ROLL, LOW PASS (under 200 ft above 350 kt), INVERTED FLIGHT and SUPERMANOEUVRE (Su-35S post-stall); SUPERSONIC, MACH 1.5 and MACH 2 once per mission; distance flown (20 XP per 10 NM, more when supersonic); and landings (Greaser +100

XP / \$300, Good +60 / \$150).

- Mission results pay out too: waves cleared, all 10 waves, 5v5 rounds and match wins, and duel wins.
- Fair levelling: each level needs a little more XP than the last (500 XP for level 2, then 150 more per level), so early levels come quickly and later ones stay reachable. Every level-up pays a cash bonus (\$250 × the new level). You never lose XP or money. Manoeuvres have cooldowns and a cap per mission so they can't be farmed, and distance doesn't count while Auto-Fly is flying the jet.
- Animations: XP and money pop up as glowing toasts as you earn them (repeats stack, e.g. LOOP ×2), a slim XP bar above the weapons fills with a glow, and a level-up sets off a full-screen celebration with light rays, expanding rings, sparks, a gold level badge slamming in, 'PROMOTED' for a new rank, the cash bonus and a fanfare. The debrief shows everything you earned, with the totals counting up and the XP bar filling through each level-up.
- Photorealistic afterburners on all four jets: each engine now has a white-hot core, a main plume with real shock diamonds and flowing turbulence, and a faint outer heat haze that fades softly at the edges instead of a solid cone. The nozzles glow white-hot in the middle and orange at the rim, and the plume stretches in thin air at altitude. Western jets burn yellow-orange fading to violet; the Su-35S keeps its blue flame with bright diamonds.
- Fixed: Su-35S gun kills were credited to the 'BK-27' (the Typhoon's cannon); they now show the GSh-30.

v1.7.6 — Smoother hangar camera

2026-09-26

- Hangar camera: the drag direction is reversed (both left/right and up/down).
- Smoother camera: moves glide with gentler easing, and a quick flick keeps the view turning for a moment before it slows to a stop. Holding still before you let go stops it dead.
- Zoom in much closer: scroll (or pinch) right up to the jet to see the cockpit, missiles and nozzles up close. The camera centres on the jet as you zoom in and never goes inside the airframe.

v1.7.5 — Hangar camera

2026-09-26

- Look around the jet in the hangar: on the main menu (and the customize screen) drag anywhere on the empty space around the jet to orbit the camera, from low beside the jet to straight overhead. Scroll the mouse wheel (or pinch on a touch screen) to zoom in close or back out. Double-click to reset the view.
- The turntable stops its slow spin as soon as you take the camera, so the jet stays where you put it.
- The loadout note on the main menu now names both missile families (the Su-35S carries R-77M and R-74M, not AIM-120D and AIM-9X).

v1.7.4 — Pusk! and model fixes

2026-09-26

- Su-35S launch call is now just 'Pusk!' (Пуск, 'launch!'), for both the R-77M and the R-74M. It's spoken in Russian if your device has a Russian voice. The F-15EX, F/A-18E/F and Typhoon keep 'Fox three' (AIM-120D) and 'Fox two' (AIM-9X).

- Model fixes for the F-15EX, F/A-18E/F and Typhoon: every wing pylon and missile rail now sits under the wing at mid-chord. Before, many hung well ahead of the leading edge; the Typhoon's outer rails started almost 3 m in front of the wingtip, and some Super Hornet pylons floated 1 m ahead of the wing.
- Stores now hang the right distance below the wing for their size: a fuel tank sits lower than a missile on the same pylon, so tanks no longer cut into the wing and missiles no longer float below it. Missiles launch from where they visibly hang.
- Landing indexer fixed for all four jets: its on-speed angle of attack now matches how each jet actually flies at approach speed (F-15EX 12, F/A-18E/F 10, Typhoon 13, Su-35S 12 degrees). Before, the F-15EX, Super Hornet and Typhoon always showed 'slow' on a correct approach.

v1.7.3 — Correct launch calls

2026-09-26

- Launch calls fixed. F-15EX, F/A-18E/F and Typhoon use the NATO brevity codes again: 'FOX 3' / 'Fox three' for the AIM-120D and 'FOX 2' / 'Fox two' for the AIM-9X.
- Su-35S launch calls are now what Russian pilots actually say. They don't use 'Fox' codes: they name the missile and call 'Пуск!' ('launch!'). The feed shows 'P-77M — ПУСК!' or 'P-74M — ПУСК!', and the voice says it in Russian if your device has a Russian voice (otherwise 'R 77 M, pusk!').

v1.7.2 — Missile calls

2026-09-26

- Launch calls now say what you actually fired. Firing an R-77M shows 'R-77M AWAY (FOX 3)' in the feed and the voice says 'R 77 M away'; an R-74M says 'R-74M AWAY (FOX 2)'. The same goes for every jet: 'AMRAAM away' for the AIM-120D and 'Sidewinder away' for the AIM-9X.

v1.7.1 — Su-35S complete

2026-09-26

- The Sukhoi Su-35S is now COMPLETE: physics, handling, looks, cockpit and weapons have all been checked against the real jet's numbers and against the other three jets.
- Performance now matches the published figures: top speed Mach 2.25 (measured 2.27), service ceiling 59,060 ft (measured 59,000), about 1,950 NM of range on internal fuel. Mach 0.9 to 1.6 at 30,000 ft in 36 s; climb to 36,000 ft in about a minute.
- Aerodynamics tuned to the Flanker: less induced and supersonic drag (it cruises and climbs high the way the real jet does) and a little more maximum lift from its big wing and leading-edge flaps. It turns with the F-15EX and Typhoon on wing alone, 18-19 deg/s sustained and about 23 deg/s instantaneous at mid speeds, and beats them all once thrust vectoring comes in at low speed.
- Top-speed limit: every jet now hits a firm barrier just past its rated top speed (the Su-35S used to creep past Mach 2.3). The other jets are unchanged.
- Supermanoeuvre mode: when the Su-35S pilot switches on the override (L), the message now reads SUPERMANOEUVRE and the HUD shows SMV · TVC, so you know the 70 deg angle-of-attack envelope and thrust vectoring are open. Switch it off and the jet goes back to its 34 deg limit.
- Takeoff and landing: the Su-35S lifts off at about 190 kt (1,600 ft on afterburner, 2,900 ft on dry power), approaches at 160 kt, and the landing indexer is tuned to its 12 deg on-speed angle of attack.

- Model fixes: the engine nacelles now slope up toward the tail like the real Flanker, so the nozzles sit at wing level and the tail clears the runway in the landing flare. Ventral fins are shorter and canted, main gear moved to match.
- Weapons now hang where they should: every wing pylon sits under the wing at mid-chord (the outer ones used to hang ahead of the leading edge), and the wingtip R-74Ms sit on the tip launch rails.
- Cockpit: the OLS-35 sensor ball no longer blocks the bottom of the HUD view (the glareshield hides it, as in the real jet). Checked both 15 in displays, the HUD, and the weapon and stores readouts for the R-77M, R-74M and GSh-30-1.
- Checked: no wobble after rolling out of turns, steady aim tracking, clean recovery from 70 deg AoA in about 1.4 s, and fair duels against every jet (roughly even with the Typhoon, a little behind the F-15EX, ahead of the Super Hornet).

v1.7.0 — Sukhoi Su-35S

2026-09-26

- New jet: SUKHOI SU-35S. Single-seat, twin-engine, super-maneuvrable air-superiority fighter: 71.9 ft long, 49 ft span, 19.4 ft tall, 76,059 lb max takeoff weight, two Saturn AL-41F1S afterburning turbofans (32,000 lbf each), Mach 2.25, 59,060 ft ceiling, 1,944 NM range, same G limits and G effects as the other jets.
- 3D thrust vectoring: the Su-35S's nozzles swivel with the controls, so it keeps full pitch, roll and yaw control at speeds where the other jets run out of air over their control surfaces. Squeeze the G-limiter override (paddle) and it can hold the nose up to 70 degrees angle of attack without departing, for Cobra-style nose pointing and very tight slow-speed turns. Release it and the jet recovers in about a second. All Su-35 pilots use it too.
- Su-35S weapons, for the Su-35S ONLY (no other jet can carry them): R-77M active-radar long-range missile (longest reach in the game, a little easier to decoy than the AIM-120D) and R-74M infrared dogfight missile (canards and thrust vectoring, slightly longer range than the AIM-9X). Keys 2 and 3 pick the IR and radar missile of whatever jet you fly. Plus the 30 mm GSh-30-1 cannon with 150 rounds.
- Su-35S sensors: N035 Irbis-E passive electronically scanned X-band radar (longest detection range in the game) and the OLS-35 optical/laserIRST for passive tracking, plus the Khibiny-M EW suite.
- Twelve hardpoints and four loadouts: Air Superiority (6x R-77M, 4x R-74M), Max Load (10x R-77M, 2x R-74M), Long Reach (8x R-77M, 2x R-74M) and Dogfight (4x R-77M, 6x R-74M), including missiles between the engines and under the intakes.
- High-detail Su-35S model: long drooped nose with the OLS-35 ball, big bubble canopy on a raised spine, blended lifting body with sharp leading-edge extensions, widely spaced engine nacelles with raked intakes, tail booms with straight vertical fins, ventral fins and stabilators, the centre tail 'sting', moving leading-edge flaps, flaperons, rudders and dorsal airbrake, and nozzles you can see swivel in flight. Blue-grey splinter camouflage with red stars, 'ВКС России' and blue or red side numbers.
- The Su-35S's afterburner burns BLUE, like the real AL-41F1S, instead of orange.
- Su-35S cockpit: two 15-inch MFI-35 wide-screen displays side by side, the PUI-35 control display and a wide-angle HUD, with its own names on the MFD pages.

- The Su-35S joins every mode: fly it yourself, or meet it as a bandit in Waves, Duel and 5v5 (bandits still never fly your own type). The TRIAD decoration now needs a kill in all four jets.
- AI jets now wear random paint jobs: your wingmen and the bandits in Waves and 5v5 get random wraps, solid colours and finishes (in 5v5 each pilot keeps the same paint all match).
- 5v5 Team Battle: the clock at the top now counts DOWN from 5:00 when the fight starts, instead of counting up. The HUD also calls out the last 60 seconds.

v1.6.0 — Black Ice

2026-09-26

- New wrap: BLACK ICE. A black nose fades into deep glacial teal toward the tail, with faceted ice crystals, smoky teal wisps and glowing cracks that shine faintly even in shadow. It's the first tile in the WRAP list; you can still change its colours, finish and brightness.
- 5v5 Team Battle: the teams now start closer together (14 NM instead of 24), so more rounds are won by shooting the other team down rather than running out the 5-minute clock.
- Faster wrap previews: making a pattern no longer freezes the customize screen for a moment.

v1.5.0 — Jet customization

2026-09-26

- Jet customization for all three jets: press CUSTOMIZE JET on the main menu to open the new customization screen, just your jet on the turntable and the paint controls.
- Paint types: the FACTORY scheme, a SOLID COLOUR (18 colours plus a custom colour picker), or a WRAP.
- Eight wraps: Digital, Splinter, Tiger, Hex, Woodland, Arctic, Carbon fibre and Chevron. Each comes with its own colours, and you can change the base and pattern colours.
- Finish: Matte, Satin, Gloss or Metallic, plus a BRIGHTNESS slider (50-150%).
- Changes preview live on the jet; nothing is saved until you press APPLY. CANCEL (or Esc) puts your saved paint back, RESET TO FACTORY starts over. Switch between the F-15EX, F/A-18E/F and Typhoon with the tabs at the top; each jet keeps its own paint.
- Your paint job is on your jet in every mode, and panel lines, roundels, tail codes and weathering stay on top of it.
- 5v5 Team Battle: a round still going after 5 minutes of fighting now ends and BOTH teams get a point (the time left shows in the HUD). If that puts both teams on the winning score together, the match is a draw.

v1.4.0 — 5v5 Team Battle

2026-09-26

- New game mode: 5v5 TEAM BATTLE. You and four AI wingmen (BLUE) against five AI bandits (RED) over Samos.
- Rounds: wipe out the other team to win the round; everyone respawns fully rearmed for the next. First team to 3 round wins takes the match (choose first to 2, 3 or 4).
- Your wingmen fly the same AI as the enemy, just on your side: they hunt, bracket, fire AMRAAMs and Sidewinders and defend themselves. Choose mixed wingman jets or all the same as yours.
- Bandits only fly the two jets you did not pick. Pick the AI difficulty (Easy to Extreme) and weapons (all, Sidewinders + gun, or guns only).

- Spectator: when you're shot down you can watch any jet on either team until the round ends. Click a jet in the list, or use the arrow keys / Tab; right-drag to orbit, wheel to zoom. It moves on to the next jet automatically when the one you're watching goes down.
- Free camera: press F while spectating to fly a camera anywhere (WASD, Q/E down/up, Shift faster, right-drag to look).
- Scoreboard: round number and score in the top bar, jets left on each side, round banners and voice calls. A round nobody finishes in 10 minutes goes to the team with more jets left (a tie replays the round).
- Logbook: 5v5 match and round record, plus two new decorations: SQUADRON LEADER (win a match) and CLEAN SWEEP (win without losing a round).

v1.3.0 — Auto-Fly

2026-09-26

- Auto-Fly replaces the old level-off autopilot. Press U to open a small panel, pick a destination (any airfield or the bullseye, or hold your current heading), a speed (300-650 kt) and an altitude (2,000-40,000 ft), then ENGAGE.
- The jet flies itself there: it turns onto course, holds your speed with the throttle (afterburner if needed), climbs over any mountains in its path, and circles overhead when it arrives.
- The destination becomes your HUD steerpoint, and the HUD shows where Auto-Fly is taking you and how far is left.
- Move the stick (or the mouse in mouse-aim) to take control back instantly; press U again to change the destination, speed or altitude, or to disengage.
- Steering fix: turns with a bank limit no longer over-pull and slowly climb (AI patrols benefit too).

v1.2.1 — Smooth roll-outs

2026-09-26

- Fixed the wobble after turning: when you stopped a turn the jet rocked wing over wing (roll one way, back, and back again). The mouse-aim autopilot now asks for a roll rate matched to what the flight controls can deliver, so the wings settle smoothly.
- Fine aim: for the last few degrees near the aim point the jet no longer swings its bank from side to side; it holds the wings steady and uses the rudder for small heading corrections.
- Unload to roll: when the jet needs to roll a long way it eases off the G first (like a real pilot), so it rolls quickly instead of fighting its angle-of-attack limit.
- Small corrections below the nose are made by easing the stick forward instead of rolling inverted.
- Gentler corrections at low speed, where the control surfaces have little authority.
- Fixed reversed rudder from v1.2.0: right rudder yaws the nose right again.
- The AI pilots use the same autopilot, so they fly smoother too.

v1.2.0 — Realistic flight physics

2026-09-26

- Flight physics rebuilt: the jets now rotate as real rigid bodies. Pitch, roll and yaw come from aerodynamic moments and the jet's inertia instead of being set directly, so every aircraft has weight, momentum and overshoot.

- Per-jet moments of inertia that change with fuel and stores: a jet loaded with wing tanks and missiles is slower to start and stop a roll.
- Real stability: pitch stability shifts aft when supersonic (less G available high and fast), the Typhoon is aerodynamically unstable like the real jet and relies on its flight-control computers, weathercock stability fades at extreme angle of attack, plus dihedral effect, adverse yaw and inertial coupling.
- Fly-by-wire modelled like modern jets: the control laws compute stabilator, aileron and rudder deflections through rate-limited actuators, with G-onset limiting (about 12 G/s). Response gets sluggish at low speed because the surfaces run out of authority, and crisp at high speed.
- Departures are possible: overriding the G-limiter at high angle of attack and low speed can stall the jet, with wing rock and nose slice; release the stick to recover.
- Engine failure yaws the jet toward the dead engine; it has to be trimmed out with rudder.
- Wind and turbulence: every mission has its own wind that strengthens and veers with height, light chop at altitude, rougher air low over land and rotor turbulence near the mountains. Gusts bump the nose and wings.
- Ground effect: the jet floats in the flare and induced drag drops near the runway.
- Control surfaces on the 3D models now show what the flight computers are actually doing (trim, damping, turn coordination).
- Fixed the F/A-18's vertical stabilizers: both now cant outward 20 degrees symmetrically (the left fin used to lean the wrong way).
- F-15EX vertical stabilizers are now perfectly straight (vertical).

v1.1.0 — High-detail aircraft

2026-09-26

- All three jets rebuilt from scratch: F-15EX Eagle II, F/A-18F Super Hornet
- Smooth blended fuselages built from real cross-sections: F-15 chines and
- Hollow intakes with rounded lips, ducts that darken with depth and engine
- Real airfoil wings and tails with rounded tips, cranked and raked planforms,
- Engine nozzles with petals, sawtooth exits, burner cans, flame holders and
- Detailed landing gear: oleo struts, torque links, drag braces, tyres and
- Pilots in ejection seats visible through the canopy, cockpit wells, glare
- Painted liveries: panel lines, rivets, walkways, NO STEP stencils,
- Speedbrakes modelled per jet: F-15 dorsal panel, Super Hornet LEX spoilers,
- New AIM-120D, AIM-9X, fuel tank, streamlined pylons and rail launchers.
- Detail parts are culled on distant jets, and each jet type is built once and

v1.0.0 — First release

2026-09-26

- Three aircraft (F-15EX, F/A-18E/F, Typhoon) over a 400 × 400 NM theater:
- Free Flight, 10-wave combat and 1v1 Duel (Easy / Medium / Hard / Extreme);
- Fly-by-wire flight model, fuel system with afterburner burn, G effects
- AIM-120D, AIM-9X, guns, flares and chaff; radar,IRST and missiles all

- AI with patrol, intercept, engage, defensive and terrain-masking behaviour.
- 3D cockpits with working displays, helmet-mounted cueing, navigation, ILS
- Pilot logbook with decorations, mission debrief, track replays, touch

Part 3 — The game today

A browser combat flight simulator in the spirit of DCS, built with TypeScript and Three.js. Six aircraft, two theaters (Frostfall Strait and Triad Isles), and an AI that uses the terrain against you.

Aircraft (the only vehicles in the game):

	F-15EX Eagle II	F/A-18E/F Super Hornet	Eurofighter Typhoon	Sukhoi Su-35S	Dassault Rafale C	Lockheed Martin F-22A Raptor
Crew	2	2	1	1	1	1
Max speed	Mach 2.5	Mach 1.8	Mach 2.0	Mach 2.25	Mach 1.8	Mach 2.25 (supercruise Mach 1.8)
Ceiling	60,000 ft	50,000 ft	55,000 ft	59,060 ft	50,000 ft	65,000 ft
Engines	2 × F110-GE-129 (29,500 lbf AB)	2 × F414-GE-400 (22,000 lbf AB)	2 × EJ200 (20,233 lbf AB)	2 × AL-41F1S, 3D thrust vectoring (32,000 lbf AB)	2 × M88-2 (16,860 lbf AB)	2 × F119-PW-100, 2D thrust vectoring (35,000 lbf AB)
MTOW	81,000 lb	66,000 lb	51,800 lb	76,059 lb	54,000 lb	83,500 lb
Gun	M61A1 20 mm	M61A2 20 mm	BK-27 27 mm	GSh-30-1 30 mm (150 rds)	30M791 30 mm (125 rds)	M61A2 20 mm (480 rds)
Radar	AN/APG-82(V) 1 AESA	AN/APG-79 AESA	CAPTOR-E + PIRATE IRST	N035 Irbis-E PESA + OLS-35 IRST	RBE2 AESA + OSF IRST, SPECTRA	AN/APG-77 AESA
Missiles	AIM-120D, AIM-9X	AIM-120D, AIM-9X	AIM-120D, AIM-9X	R-77M, R-74M (Su-35S only)	Meteor, MICA IR (Rafale only)	AIM-120D, AIM-9X (internal bays)

Current version: **v4.5.1** — see CHANGELOG.md (also in the game under **v4.5.1 • NOTES** on the main menu).

Each jet is a high-detail procedural model: about 250k-390k triangles for the jets around you, and a hero build with about 10x that (2.5-4 million triangles) for your own jet and the hangar: blended fuselages

built from real cross-sections, hollow intakes with ducts and fans, airfoil wings and tails with moving control surfaces, petal nozzles with burner cans, detailed landing gear, seated pilots and painted liveries with panel lines, rivets, stencils and weathering.

Missiles: AIM-120D AMRAAM (active radar, datalink midcourse, loft) and AIM-9X

(IR, high off-boresight) on the F-15EX, Super Hornet and Typhoon; R-77M (active radar, dual-pulse) and R-74M (IR, canards + thrust vectoring) on the Su-35S only; Meteor (ramjet, about 92 NM reach and a 34 NM no-escape zone, the longest in the game) and MICA IR (imaging IR, thrust vectoring, about 27 NM) on the Rafale only. Countermeasures: flares and chaff.

Playing

```
npm install
npm run dev # http://localhost:5173
```

Other builds:

```
npm run build # multi-file site in dist/ (what GitHub Pages serves)
npm run build:single # ONE self-contained HTML file in dist-single/index.html
npm run typecheck
```

dist-single/index.html has no external dependencies (fonts fall back to system fonts offline); you can double-click it and play from disk.

Game modes

- **Free Flight** — pick a jet and a friendly airbase (runway or air start). No enemies.
 - **Waves** — ten escalating waves. Waves 1-3: 3 basic jets. 4-6: 6 tactical jets that defend properly and terrain-mask. 7-9: 6 aggressive jets with mid-range AIM-120D shots. Wave 10: 9 elite jets with coordinated multi-ship targeting.
 - **1v1 Duel** — choose your jet, your opponent (one of the two you did not pick) and the difficulty (Easy / Medium / Hard / Extreme). Start head-on over the contested island (Hvitøya or Samos), or on its opposite runways with the mountain between you. Rules: all weapons, Sidewinders + guns, or guns only.
Enemy AI is never the same type as the player's jet.
 - **5v5 Team Battle** — you and four AI wingmen against five AI bandits over the contested island. Wipe out the other team to take the round; everyone respawns; first to 3 wins.
When you're shot down, spectate any jet on either team or fly a free camera (F).
A round still going after 5 minutes ends with a point to both teams.
 - **Free-for-all (Last Pilot Standing)** — 12 jets, you and 11 AI, everyone hostile to everyone, no respawns. The battle zone shrinks in stages and the storm outside it destroys jets that stay out. Kills refill a missile of each type; the top scorer carries a revealed bounty; the last two fight a FINAL DUEL. Your placing out of 12 goes on the scoreboard.
- Customize** (main menu): solid colours, 12 wrap patterns (including Black Ice, Inferno, Aurora and Galaxy), finish and brightness for each jet.

Graphics

SETTINGS → GRAPHICS has overall presets (Low, Medium, High, Ultra, 4K Ultra) and every option on its own: render resolution up to 4K (3840 × 2160) with a resolution scale, MSAA up to 8×, shadow quality, mountain lighting (sun shadows cast by the terrain across the whole map and sky occlusion in valleys), light scattering, bloom, tone mapping (Neutral / Filmic / AgX), brightness, contrast, saturation, vignette, world detail, cloud amount (up to overcast), cloud quality and drifting cloud shadows.

The theaters

Pick the map on the **THEATER** card on the main menu. Every mode works on both.

Frostfall Strait (default): 200 × 200 NM frozen arctic archipelago, with snow-covered mountains (huge Fuji-style cones, the tallest over 26,000 ft), glaciers, ice cliffs, pack ice and still, depth-shaded water.

Fights start around 29,500 ft.

- **Nordland** and **Sørvik** (blue) — Nordhavn AB, Isvik AB.
- **Østmark** and **Kragfjell** (red) — Kragen AB, Svalbru AB.
- **Hvitøy** (contested) — a wall of peaks between Hvitøy West AB (blue) and Hvitøy East AB (red).

Triad Isles: 400 × 400 NM of sea with three islands, all covered in forest and hills (no grass):

- **Skye** (blue) — steep mountains, sea lochs, coastal cliffs. Dunvegan AB, Broadford AB.
- **Capri** (red) — compact, rocky limestone crags, blue grottoes. Anacapri AB, Marina Grande AB.
- **Samos** (split) — high pine-covered hills and pebble beaches, with a great N-S ridge. Karlovasi AB (blue, west) and Vathy AB (red, east) cannot see each other.

Radar, IRST, ground radars and missile seekers all need real line of sight:

terrain and the curvature of the earth block them, so you can hide in sea lochs or behind the Samos ridge, and so can the AI.

Controls (defaults, rebindable in CONTROLS)

Key	Action	Key	Action
Mouse	Aim (mouse-aim mode)	LMB / Space	Fire
W / S	Pitch down / up	A / D	Roll
Q / E	Rudder	Shift / Z	Throttle up / down
Tab	Afterburner	1 / 2 / 3	Gun / AIM-9X / AIM-120D
R / T	Lock / unlock	Y	Radar mode
I	IRST (Typhoon)	C / V	Flares / chaff
G	Gear	B	Speedbrake
N	Wheel brakes	L	G-limiter override

Key	Action	Key	Action
F	Cockpit / chase camera	F1-F4, F6	Cockpit, chase, fly-by, target, weapon cams
RMB drag	Look around	[/]	Zoom
M	Theater map	H	Rearm & refuel (stopped at a friendly base)
K	Jettison tanks	J (hold)	Eject
U	Auto-Fly (destination, speed, altitude)	Esc / P	Pause
, / .	Left / centre / right display page	\	Cockpit cursor (click displays)
'	Next steerpoint	End	Steer to nearest friendly field

Gamepads are supported. Mouse modes (mouse-aim, virtual joystick, keyboard-only) are in SETTINGS.

Cockpits and avionics

Each jet has its own 3D cockpit with working displays:

- **F-15EX** — a 10×19 in touch-screen large-area display split into two portals, up-front controller, standby display, caution panel, LOCK/SHOOT lights.
- **F/A-18E/F** — two DDIs, the up-front controller display, 8×8 in centre MFD, engine/fuel display and standby instruments.
- **Typhoon** — three colour MHDDs, the dedicated warning panel and the get-u-home standby display.
- **Su-35S** — two 15 in MFI-35 wide-screen displays side by side, the PUI-35 control display and a wide-angle HUD.
- **Rafale** — a head-level display between two lateral displays, and the wide HUD.

Display pages: **RADAR** (B-scope, RWS/TWS/ACM/silent, track files, STT data block, launch zone, jammer strobes, why a lock was lost), **TSD / SA / PA** (heading-up moving map with coastlines, contours and the datalinked picture), **STORES, ENGINE, FUEL** (bingo/joker, fuel at steerpoint), **EW** (threats, jammer, countermeasure program), **HSI** (TACAN, ILS) and **FCS/DAMAGE**. Click the bezel buttons with the cockpit cursor (\) or cycle pages with , / ..

Helmet-mounted cueing (JHMCS / JHMCS II / Striker II): look at a target in the cockpit and press **R** to lock it; the AIM-9X seeker follows your head.

Navigation: steerpoints for every field plus bullseye (' next, End = nearest friendly field), HUD steerpoint cue, ILS on every runway (approach valleys keep the glideslope clear of the mountains), an AoA bracket, and graded touchdowns.

Logbook, debrief and replay

Every sortie is logged: kills by weapon and range, missiles defeated, landings, G / Mach / altitude records, duel record by difficulty, best wave and 18 decorations (LOGBOOK on the main menu). The results screen shows an engagement map and timeline, and **WATCH REPLAY** plays the whole mission back with scrubbing, 0.25–8× speed and chase / fly-by / target cameras on any aircraft.

Phones and tablets

Touch controls (virtual stick, throttle slider, fire/lock/weapon/countermeasure buttons, drag to look) switch on automatically on touch devices.

Flight and physiology

- Six-degree-of-freedom physics: forces *and* moments. Each jet has its own moments of inertia (changing with fuel and wing stores), stability and damping derivatives, and control power; rotation follows Euler's equations.
- Fly-by-wire: the stick commands load factor and roll rate; the control laws work out stabilator / aileron / rudder deflections (rate-limited actuators, G-onset limiting, AoA and G limiters; override with L). Authority fades at low speed, stability stiffens supersonic, the Typhoon is unstable without its computers, and pushing past the limits can depart the jet.
- Wind and turbulence (per-mission wind, chop aloft, rough air low over land, mountain rotor), ground effect, and asymmetric thrust with an engine out.
- Fuel matters. Afterburner multiplies fuel flow several times over.
- G effects: +4.0 to +7.9 G grey-out; +8.0 to +10.4 G tunnel vision; +10.5 G and above G-LOC (10 s blackout, controls frozen). –2.0 to –4.9 G red-out at 50 %; –5.0 G and below full red-out.

Code layout

```
src/core math, noise, atmosphere, input, settings
src/world terrain function, islands, LOD streaming, trees, ocean, clouds, airfields
src/render renderer, fog, sky, particles, G-effect post-processing, cameras
src/aircraft specs, flight model, damage, procedural 3D models (models/kit.ts: loft, wing, livery shader), cockpit
src/weapons AIM-120D / AIM-9X, guns, countermeasures
src/sensors radar,IRST, RWR / MAWS, signatures
src/ai skill levels, steering controller, AI pilot state machine
src/game simulation, team air picture / GCI, spawning, game modes, logbook, replay
```

src/avionics display pages, navigation (steerpoints, TACAN, ILS, fuel planning)
src/ui HUD, helmet display, scopes, map, menus, hangar, debrief, touch and replay controls
src/audio synthesized engine, weapon and warning audio

Deploying

.github/workflows/deploy.yml builds and publishes to GitHub Pages on every push to main (enable Pages with source "GitHub Actions" in the repository settings).

Multiplayer

MULTIPLAYER ► on the main menu opens the server browser: the official servers **OFFICIAL 1-5** with live pilot counts, or any server by address.

Online play is LAST PILOT STANDING between real players only — there is no AI on any server. Between matches everyone flies with weapons on hold; a match starts when two or more pilots are in (15 s countdown), everyone drops in on a ring around the arena, the zone shrinks, kills rearm you, the last jet flying wins, and the next match starts by itself. Pilots who join mid-match spectate until the next one. Each room is up to 12 pilots and has a fixed theater (the game loads it when you join).

Play online from the website build (GitHub Pages): browsers only allow secure wss:// connections from an https:// page.

Official servers (OFFICIAL 1-5)

One server process hosts all five rooms (OFFICIAL 1-3 on Triad Isles, OFFICIAL 4-5 on Frostfall Strait):

```
cd server && npm install && node server.mjs --official
```

render.yaml deploys exactly that on Render (New → Blueprint → this repo), which gives wss://triad-servers.onrender.com — the address the game looks for (src/net/servers.ts; a build can point elsewhere with the VITE_TRIAD_SERVER environment variable). Any Node 18+ host or server/Dockerfile works too; it listens on \$PORT (default 8080) and GET /status lists the rooms.

Host your own

```
cd server && npm install
```

```
node server.mjs --name "MY SERVER" --map triad --port 8080 # or --map frost, --max 8
```

Friends join with **JOIN A SERVER BY ADDRESS**. Over the internet the server

needs a secure address (a free Cloudflare Tunnel, Render, or a TLS reverse proxy); on your own network `192.168.x.x:8080` works from the offline build (`triad-offline.html`, opened from disk).

How it works

Each client flies its own jet and its own weapons; the server relays jet state (20 Hz), missiles, countermeasures and hits, and runs the match and the zone. What the shooter sees decides a hit (its gun rounds and missiles hit the other jet on its screen), the hit is sent to that pilot, and their client takes the damage and reports the kill. Other jets are shown 0.12 s behind real time and interpolated so they move smoothly.

Part 4 — Every commit

Date	Commit	Message
2026-09-24	db452a9	Initial commit
2026-09-26	d0bd486	Add simulation core: terrain, world rendering, flight model, weapons, sensors, AI
2026-09-26	c7f4e95	Add game shell, modes, HUD, menus, cockpit, effects and audio
2026-09-26	6f29618	Make terrain worker fallback self-driving; use classic workers
2026-09-26	d795a85	Add README and GitHub Pages build/deploy workflow
2026-09-26	26b1b38	Add per-aircraft 3D cockpits and a multi-page avionics suite
2026-09-26	0839d02	Refresh camera matrices before drawing the HUD; clip HUD to the combiner
2026-09-26	33542a2	Add helmet-mounted cueing (JHMCS / JHMCS II / Striker II)
2026-09-26	1dce8bf	Navigation and landing aids: HUD steerpoint/ILS/AoA cues, approach corridors, touchdown grading
2026-09-26	bbae1be	Add pilot logbook, sortie debrief and decorations
2026-09-26	07ea72d	Add ejection sequence, parachutes and airframe debris
2026-09-26	30a98d6	Add touch controls for phones and tablets
2026-09-26	3686c2b	Add track recording and a replay viewer
2026-09-26	ce3e2a8	README: document cockpits, avionics, navigation, logbook, replay and touch controls
2026-09-26	68038fa	Warm Capri's limestone so terrace risers no longer read as snow
2026-09-26	34fabd2	Show friendly aircraft in green on radar, minimap, MFDs and the theater map
2026-09-26	2714222	v1.1.0: rebuild all three jets as high-detail models, add version notes
2026-09-26	575e981	Remove a stray test screenshot
2026-09-26	1cdb1e8	v1.2.0: six-degree-of-freedom flight physics, wind; fix fin cant
2026-09-26	f57ec3e	v1.2.1: stop the wing-over-wing rocking after turns; fix rudder sign
2026-09-26	941947a	v1.3.0: Auto-Fly replaces the level-off autopilot
2026-09-26	e4ffa71	v1.4.0: 5v5 Team Battle mode with spectator and free camera
2026-09-26	b1a7d8e	v1.5.0: jet customization screen; 5-minute team rounds score for both
2026-09-26	1ceebb3	v1.6.0: Black Ice wrap, closer 5v5 spawns
2026-09-26	bd10979	v1.7.0: Sukhoi Su-35S, thrust vectoring, R-77M/R-74M, AI paint, 5v5 countdown
2026-09-26	7cdf092	v1.7.1: Su-35S complete — performance, handling, geometry and cockpit polish
2026-09-26	32962b5	v1.7.2: launch calls name the missile fired
2026-09-26	9536299	v1.7.3: correct launch calls (FOX 3 / FOX 2; "Пуск!" for the Su-35S)
2026-09-26	74cbebe	v1.7.4: "Пуск!" call, wing stores seated under the wings, indexer AoA
2026-09-26	7e1ab0e	v1.7.5: free orbit camera in the hangar
2026-09-26	b81aa21	v1.7.6: reversed, smoother hangar orbit with close zoom
2026-09-26	0b32fbe	v1.8.0: pilot XP / levels / money, photorealistic afterburners
2026-09-26	8720fcb	WIP: second map "Frostfall Strait" (not finished)
2026-09-26	883f103	v1.9.0: Frostfall Strait map, still depth-shaded water, 3 new wraps, fixes

Date	Commit	Message
2026-09-26	f8e9b54	v1.9.1: broad Fuji-style giant mountains on Frostfall Strait
2026-09-26	bd6a5d3	v1.9.2: chase camera recenters smoothly after 1.8 s idle
2026-09-26	084603f	v2.0.0: graphics overhaul
2026-09-26	60d1d2c	v2.1.0: Free-for-all mode (Last Pilot Standing)
2026-09-26	9205fbb	v2.1.1: fix black screen in flight (sanitize HDR buffer before bloom)
2026-09-26	90614b5	v2.2.0: remove the pilot XP / level / money system
2026-09-26	5aa55d5	v2.2.1: twin-rail missile racks, aligned on all jets
2026-09-26	711da7e	v2.2.2: fix multi-second freeze when a jet is destroyed
2026-09-26	1f1926e	v2.2.3: missile racks matched to the real Super Hornet, Typhoon and Su-35S
2026-09-26	920799b	New sound engine: realistic jet engine, airflow, weapons and warnings
2026-09-27	eda178d	v3.0.0: new sound engine and more realistic jet models
2026-09-27	19ace6c	v3.1.0: physics update and bug fixes
2026-09-27	eabe756	v4.0.0: multiplayer (official servers 1-5, self-hosted servers)
2026-09-27	879dad3	v4.0.1: Su-35S and Typhoon intakes raked the right way up
2026-09-27	dd98bd4	v4.1.0: the Dassault Rafale C arrives
2026-09-27	a464b7b	v4.2.0: hero-detail jet models (~10x triangles)
2026-09-27	31db377	v4.3.0: weather (clear, cloudy, overcast, rain, storm, snow) and new clouds
2026-09-27	f412ceb	v4.3.1: rain droplets on the screen, no more cloud popping
2026-09-27	a2d59eb	v4.3.2: smaller rain drops, only at the screen edges
2026-09-27	348d64f	v4.3.2: release notes and version
2026-09-27	79dc52f	v4.4.0: War Thunder style gun lead marker
2026-09-27	aef6a79	v4.4.1: cloud sort stutter, weather reflections, weather panel on small screens
2026-09-27	4128564	v4.5.0: F-22A Raptor (supercruise, 2D thrust vectoring, internal bays, hero model)
2026-09-27	82ac87a	v4.5.1: F-22A model rebuilt (faceted body, integrated caret intakes, tail booms, 2D nozzles)